



Cactus

Chemometrics and Analytical Chemistry Tools

User Guide

A user-friendly software for exploring, preprocessing and modelling analytical data
through chemometric and machine-learning workflows
without writing code

Version 1.0.0

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This guide describes the application from the perspective of the end user. It does not document internal source code, algorithms, or implementation details.

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Chapter 1

Introduction

1.1 What Cactus Is

Cactus is a software application for the analysis of tabular analytical data using chemometric and machine-learning methods. It is designed primarily for numeric and high-dimensional datasets, such as spectroscopic data containing many predictor variables, but it can also be used with more general classification and regression datasets.

The application provides a guided workflow covering the main stages of data analysis:

1. **Import** data from CSV, TXT or Excel files.
2. **Edit** the dataset by selecting samples and variables, assigning variable roles, managing missing values and defining Training and Test sets.
3. **Explore** the data using graphical tools and statistical tests.
4. **Preprocess** predictor and outcome variables using common chemometric transformations.
5. **Model** the data using unsupervised, classification, one-class and regression methods.
6. **Predict** new samples using a previously saved model.

Cactus is built with R and Shiny, but no knowledge of R or programming is required to use the application.

1.2 Who This Guide Is For

This guide is intended for anyone who wants to explore, analyse and model their data without writing code.

It describes the main application workflow, the available controls and the practical meaning of the most important settings. Statistical and chemometric concepts are introduced only where they are necessary for selecting and using a method correctly.

1.3 Supported Data Structure

Cactus expects the dataset to be organised as a table in which each row represents a sample or observation and each column represents a variable.

```

mpc1248,0714555,0.151574,1.4215,0268,005,0.421,3417,009,0.431,0656,007,0.427,005,000,0.426,043,
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```

(b) Example of csv/text structured dataset

During the Edit step, columns can be assigned one of the following roles:

- **Predictor (X)** – Numeric variables used as model inputs, such as spectral intensities, chemical measurements or instrumental signals.
- **Classification outcome (Y)** – Categorical variables containing class or group labels.
- **Regression outcome (Y)** – Numeric variables used as response values in regression models.
- **Date/time** – Variables used to order samples, create time-based splits or produce temporal visualizations.

A dataset may contain multiple classification and regression outcomes. The variable to be analysed is selected within the relevant modelling module.

The main sections of Cactus follow the order shown below:

Stage	Main functions
Import	Load the original dataset
Edit	Modify data, assign roles, manage missing values and split samples
Explore	Visualizations and statistical analyses
Preprocess	Transform predictor and outcome variables
Model	Unsupervised, classification, one-class and regression methods
Predict	Apply a saved model to new data

The workflow is sequential. Sections that require processed or approved data remain unavailable until the necessary previous steps have been completed.

If a section is hidden or disabled, check that the preceding stage has been completed and confirmed. This behaviour prevents analyses from being run on incomplete or unapproved data.

Chapter 2

Installation and Launching the Application

Cactus runs locally on Windows using a bundled copy of R. No separate installation of R or of any R package is required if the bundled runtime folder is present.

2.1 What you need

- The application folder, which must contain (among others):
 - `app.R`, `global.R`, `ui.R`, `server.R` – the application itself.
 - `R-4.5.2\` – a self-contained R installation used to run the app.
 - `start_app.bat` – the Windows launcher script.
- A modern web browser (Chrome, Edge or Firefox). Cactus opens inside your default browser; it is not a separate desktop window.

2.2 Starting the application on Windows

1. Double-click `start_app.bat` inside the application folder.
2. A console window opens and prints the R startup and package-loading progress (`[CACTUS ...s] ...loaded`), followed by `[CACTUS3] Starting on port 3838` – the console still prints the application’s internal build tag; this is expected and not an error.
3. Once loading finishes, your default browser opens automatically at `http://127.0.0.1:3838` (or the port shown in the console) and the application appears, with a short “Loading CACTUS...” splash screen.
4. Keep the console window open while you use the application – closing it stops the app.

💡 Tip

The launcher accepts two optional command-line arguments: a port number and a secret key (`start_app.bat [port] [secret_key]`). If you were given a specific port to use by your administrator, run the script from a command prompt with that port as the first argument, e.g. `start_app.bat 4000`.

**Note**

If a browser tab does not open automatically, open it manually and navigate to the address shown in the console window (typically `http://127.0.0.1:3838`).

2.3 Closing the application

Use the **Logout** button in the right-hand settings panel (see Section 3.3), or simply close the console window that opened when you started `start_app.bat`.

2.4 First-time setup / package installation

The console comments in `start_app.bat` reference separate one-time install scripts (`install_step1.bat` for core and Import packages, `install_step6.bat` for the unsupervised-modelling package `mdatools`) intended to be run once, in order, before the first launch, if the bundled R installation does not already have all required packages.

TODO

The `install_step*.bat` scripts referenced in the comments of `start_app.bat` were not found in the current repository. If your copy of Cactus does not start correctly because of a missing R package, contact your administrator or the development team (Chapter 15) rather than attempting to install packages manually.

2.5 Server / multi-user deployment

`global.R` and `server.R` reference an optional "launcher mode" (`SPECTRAPP_LAUNCHER_MODE`) that integrates Cactus with a separate login application. This mode is intended for shared/server deployments where multiple users log in before reaching Cactus, and is configured by whoever deploys the server.

TODO

The exact setup of the external login application and shared-server deployment is outside the scope of this user guide and not fully derivable from the application code alone. If you are accessing Cactus through a shared server rather than running `start_app.bat` yourself, ask your administrator for the correct URL and login instructions.

Chapter 3

Application Overview

3.1 Main layout

Cactus uses a classic dashboard layout with three regions:

- **Header (top bar)** - shows the Cactus logo, a **theme toggle** button (sun/moon icon) that switches between light and dark appearance, and a **gear icon** that opens the right-hand settings panel.
- **Sidebar (left)** - the main navigation menu. Each entry corresponds to one stage of the analysis pipeline described in Chapter 1.
- **Body (centre)** - the content of the currently selected sidebar tab: forms, tables and plots.

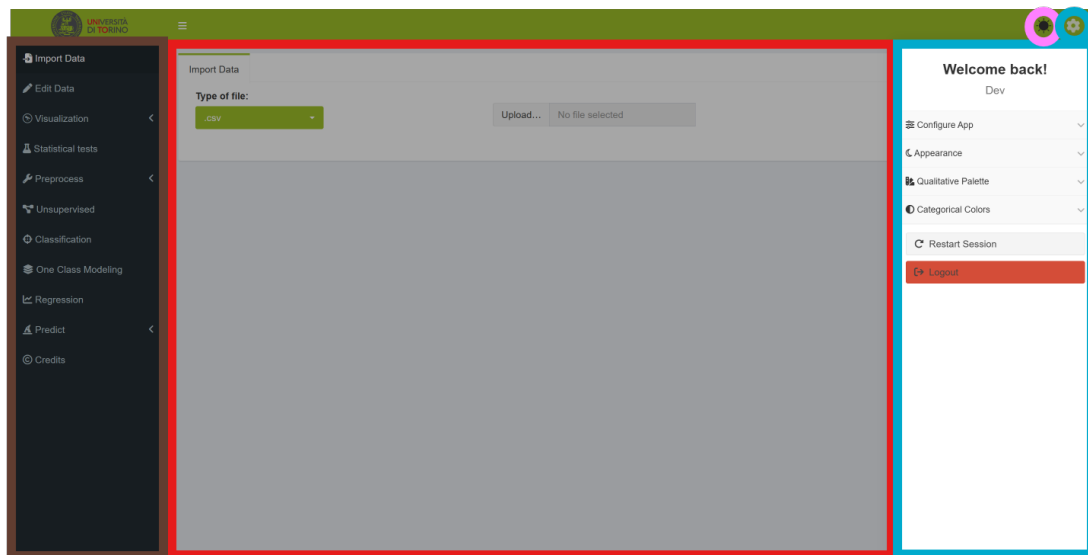


Figure 3.1: Main application layout (header, sidebar, body)

3.2 Sidebar navigation

The left sidebar lists, from top to bottom:

1. **Import Data**
2. **Edit Data**
3. **Visualization** (One Variable / Two Variables / Multiple Variables)
4. **Statistical Tests**
5. **Preprocess** (Predictors / Outcome / Data Preprocessed)
6. **Unsupervised**
7. **Classification**
8. **One Class Modeling**
9. **Regression**
10. **Predict** (Upload Model / Prediction)
11. **Credits**

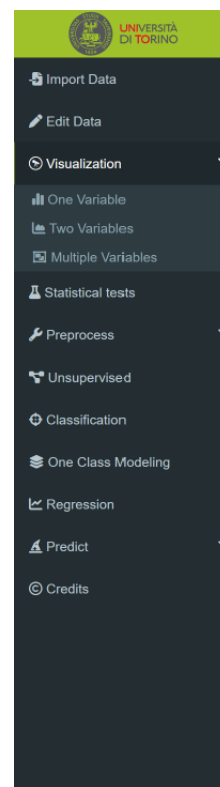


Figure 3.2: Left sidebar navigation with all sections unlocked.

Warning

Several sidebar entries are **gated**, meaning that they only appear or become available after the required previous steps have been completed.

- The **Preprocess** submenu only appears after clicking **Approve Dataset** in the **Edited Data** tab.
- **Unsupervised**, **Classification**, **One Class Modeling**, and **Regression** become available only after preprocessing has been completed.

If a section is empty or displays a message asking you to complete a previous step, return to the required module and finish that step first.

3.3 Right-hand settings panel

Clicking the gear icon in the header opens a panel on the right with several collapsible sections:

- **Qualitative Palette** - the default colour palette used for continuous/gradient plots (choose any ColorBrewer palette, a saved custom palette, or build a new one via "Create your Palette").
- **Categorical Colors** - eight default colours used for categorical/factor variables across all plots; click **Edit Colors** to change them.
- **Appearance** - a **Dark mode** switch, saved as your default for future sessions.

- **Configure App** - lets you show or hide individual optional tabs and options throughout the app (e.g. which chart types appear under Visualization, which cross-validation methods are offered, which metrics are shown in classification/ regression/SIMCA results, and the PLS-DA classification rule). Hiding a tab here does not delete any data; it only declutters the interface.
- **Restart Session / Logout** - at the bottom of the panel.

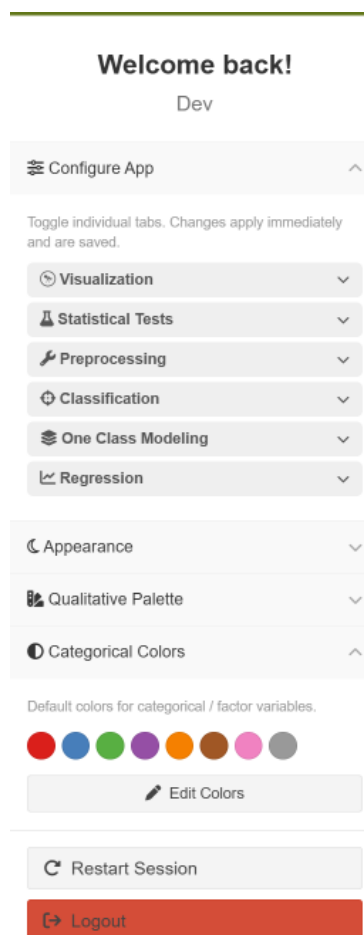


Figure 3.3: Right-hand settings panel (Appearance and Configure App sections open).

Tip

If an expected chart type, metric, or cross-validation method seems to be missing from a module, check **Configure App** in this panel - it may simply be switched off.

3.4 Common UI elements

These elements recur throughout the application:

- **Boxes** - most content is grouped into titled boxes. A box with a green header is a "success"-styled box, typically used for confirmed/completed steps.
- **Maximize button** (top-right corner of a box, expand icon) - enlarges that box (and any plot/table inside it) to fill the screen; click again to restore.

- **Info button** (circled "i" icon) - opens a short explanation or a References modal for that box's content.
- **Settings / gear dropdown** - above many tables and plots; controls how many rows to display, plot colours, labels, and similar cosmetic options.
- **Download / Export buttons** - appear next to tables (CSV/Excel export) and plots (image export); see Chapter 16 and Chapter 18.
- **Confirmation pop-ups** - destructive or pipeline-changing actions (removing columns, resetting a pipeline, approving a dataset) always ask for confirmation before proceeding.

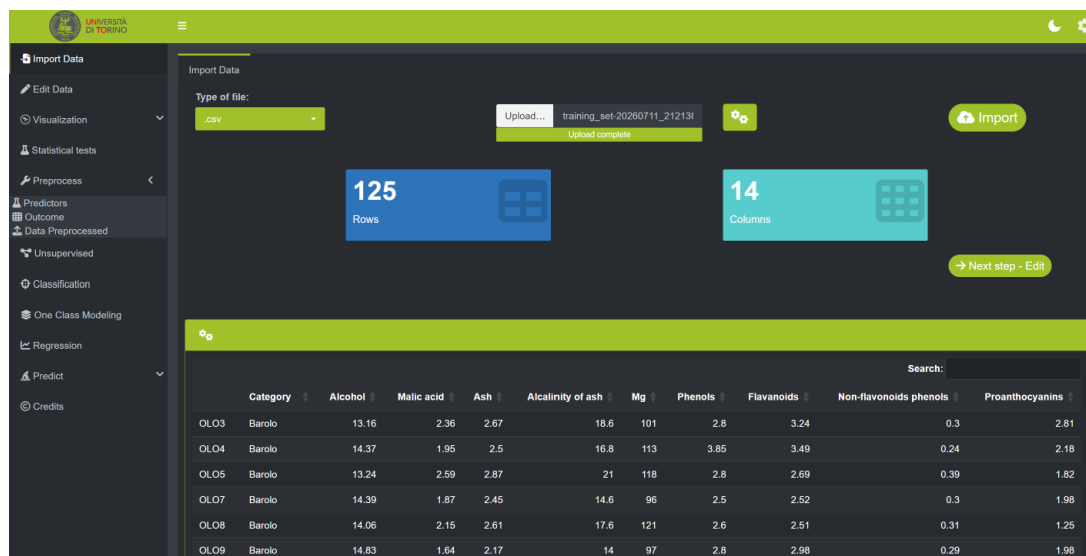


Figure 3.4: Dark theme

3.5 Moving from import to analysis to export

The intended path through the application is linear:

1. Import your dataset (Chapter 5).
2. Clean and configure it in Edit Data, then click **Approve Dataset** (Chapter 6).
3. Optionally explore it visually and statistically (Chapters 7 and 8).
4. Preprocess the predictor and/or outcome variables (Chapter 9).
5. Run one or more modelling techniques appropriate to your goal: unsupervised exploration (Chapter 10), classification (Chapter 11), one-class modelling (Chapter 12) or regression (Chapter 13).
6. Save a trained model, then later reuse it on new data via Chapter 14.

Any change to the Edit step (re-approving the dataset, changing which variables are predictors) automatically resets downstream models, so that results are never left inconsistent with the data used to produce them.

Chapter 4

Import and Data Management – Overview

This chapter gives a high-level view of how data enters Cactus and how it is prepared for analysis. Full, step-by-step instructions for each screen are given in Chapter 5 (Import module) and Chapter 6 (Edit module).

4.1 Getting data into the application

Data is loaded on the **Import Data** tab by uploading a single file in one of three formats: CSV, TXT (delimited text) or Excel (`.xls/.xlsx`). The file should contain one row per sample and one column per variable, with column headers on the first row (unless you explicitly indicate otherwise). Advanced options let you control the delimiter, decimal symbol, Excel sheet number, whether the first column holds row labels, and whether to automatically drop empty or constant columns.

4.2 Expected data format

- **One table, one dataset:** all variables (predictors, class labels, regression targets, dates) must be columns of the same file.
- **Row labels:** if the first column contains unique sample identifiers, enable "Use first column as row labels" so it is not treated as a data variable. Duplicate values in that column will block the import.
- **Column types:** Cactus automatically detects numeric, categorical and date/time columns after import. Numeric columns with text mixed in (e.g. stray units) will not be recognised as numeric.

4.3 Class and response columns

Cactus does not require you to designate class or response columns at import time. Instead, once the data is imported you assign the *role* of every column during the Edit Pipeline step:

- **Classification variables** – categorical columns used as class labels (Classification/Discrimination and One Class Modeling).
- **Regression variables** – numeric columns used as a modelling target (Regression).
- **Date variables** – optional date/time columns.

- **Predictor (X) variables** – everything else that should be used as model input; by default all remaining numeric columns.

Any column left unassigned at the end of the Edit Pipeline must be explicitly deleted before you can proceed – Cactus will not silently guess its role.

4.4 Approving the dataset

Once columns are typed and assigned, and any missing values are resolved, the **Edited Data** tab shows the final preview and an **Approve Dataset** button. Approving the dataset is the point of no return that unlocks Visualization, Statistical tests, Preprocessing and all modelling tabs; changing your mind afterwards means editing again and re-approving, which resets any models you had already trained.

4.5 Common import errors

- **File type mismatch** – the uploaded file’s extension does not match the selected ”Type of file”; choose the matching type or file.
- **Duplicate row labels** – occurs when ”Use first column as row labels” is enabled but that column has repeated values; either uncheck the option or make the identifiers unique before importing.
- **Date parsing warnings** – shown when a column looks date-like but could not be parsed reliably; check the source formatting of that column.

See Chapter 5 for the complete, screen-by-screen reference, and Chapter 18 for a consolidated list of error messages and fixes.

Chapter 5

Import

The Import module (sidebar entry **Import Data**) is the first screen you see: it loads your data file into Cactus so it can be cleaned, explored and modelled in the next steps. Follow the steps below in order.

5.1 Step 1 - Choose the file type

Open the **Type of file** dropdown at the top of the page and select the format of your dataset: **.csv**, **.txt**, or **.xls/.xlsx**. Pick the option that matches your file's real extension - Cactus uses it to decide how to read the file.

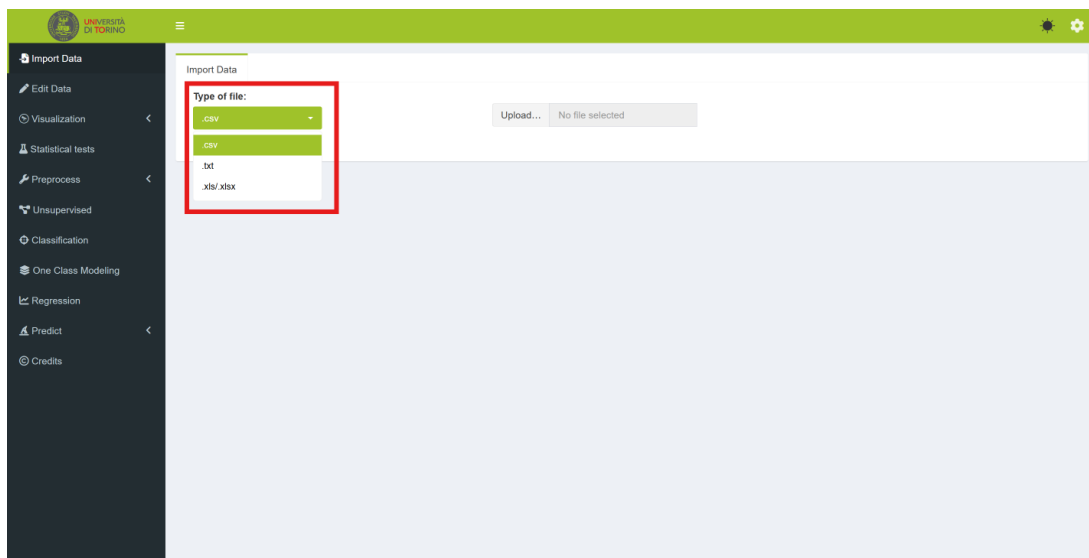


Figure 5.1: Type of file dropdown

5.2 Step 2 - Upload your file

Click **Upload...** The file browser that opens will only let you pick files matching the type you selected in Step 1 - if your file does not appear in the list, go back and check that you chose the correct extension.

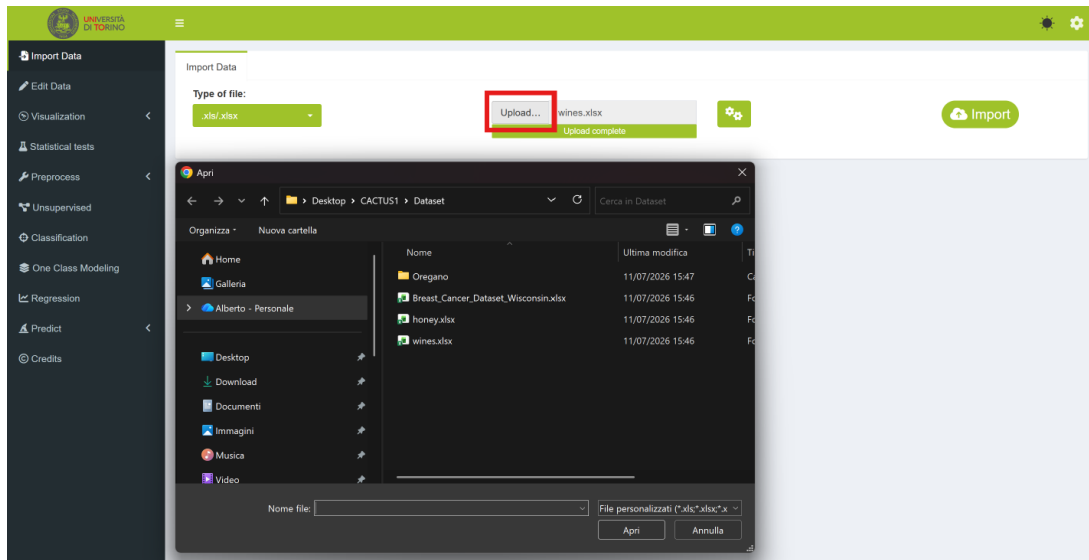


Figure 5.2: Upload button and file browser filtered to the chosen file type

5.3 Step 3 - Adjust advanced options (optional)

If your file needs special handling, click the gear icon to open **Advanced options**:

- **Sheet number** - which worksheet to import (Excel files only, default: 1).
- **Delimiter** - the column separator (CSV/TXT only; default comma for CSV, tab for TXT; leave blank to auto-detect).
- **Decimal symbol** - the character used for decimals (default: . - change this to a comma if your file was exported with European settings).
- **Use first column as row labels** - turn this on if your first column holds sample names/IDs rather than data (default: on).
- **Remove constant columns** - drops columns where every row has the same value (default: off).
- **First row is data (no header)** - turn this on if your file has no column-name row; columns are then auto-named `var_1`, `var_2`, ... (default: off).
- **Remove empty columns** - drops columns that are entirely empty (default: on).

Most files need no changes here - only open this panel if Step 4 fails or the preview looks wrong.

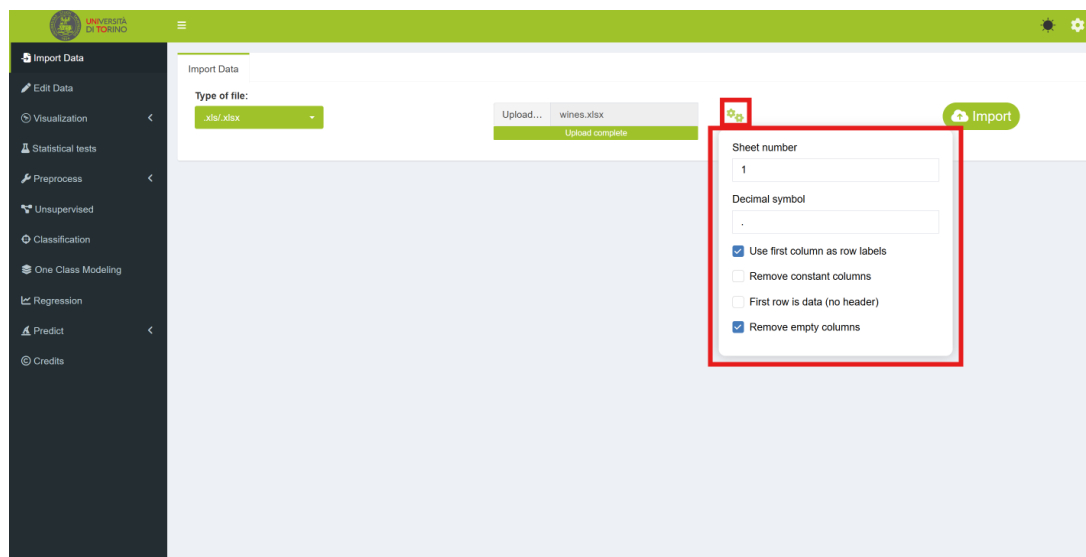


Figure 5.3: Advanced options dropdown open

5.4 Step 4 - Click Import

Click the green **Import** button. A pop-up confirms how many rows and columns were imported, or explains what went wrong (see Common mistakes below).

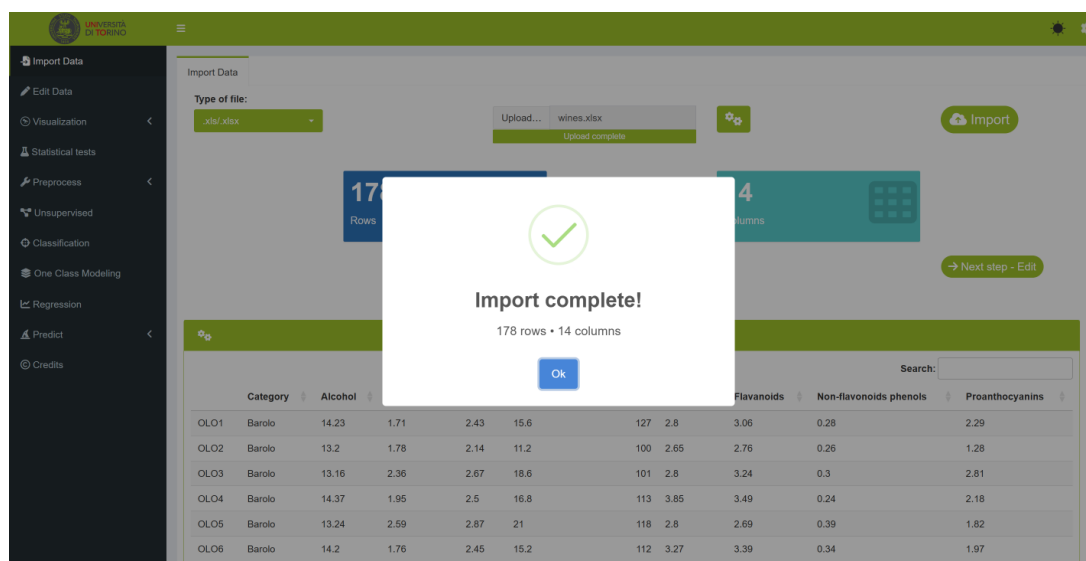


Figure 5.4: Import success pop-up with row/column counts

5.5 Step 5 - Check the preview table

Scroll through the preview table to confirm the data looks right: column names, numeric values, and dates should all appear as expected. Use the gear-icon settings dropdown above the table to show more rows/columns or to download the imported data as CSV.

Warning

Displaying the entire dataset on screen by selecting the **Show entire dataset** checkbox may significantly slow down the application. It is therefore recommended to avoid this option. This table is intended only as a preview to verify that the data import was successful.

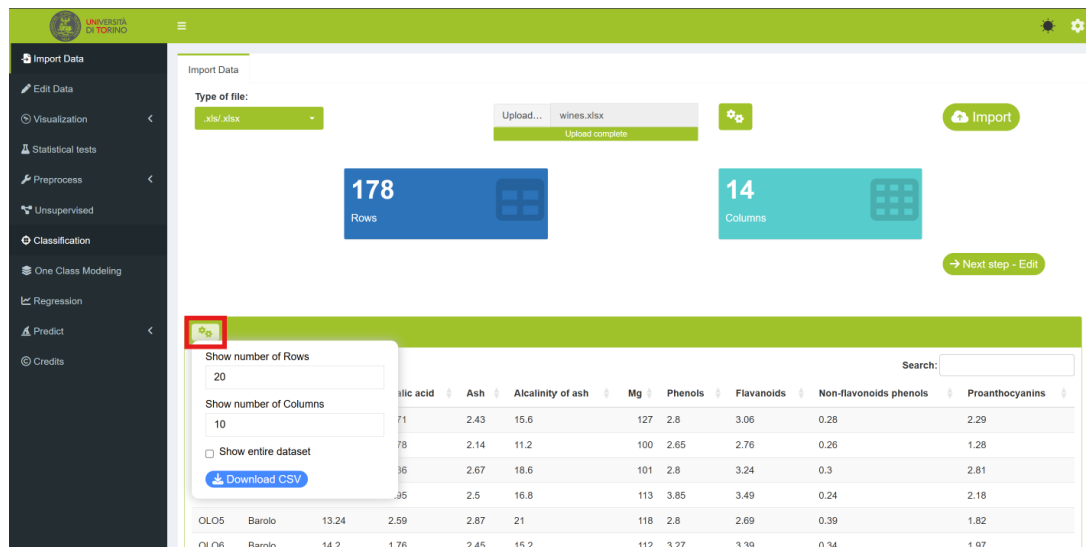


Figure 5.5: Data preview table with row/column count boxes

Tip

Cactus automatically tries to detect date/time columns. A successful import can generally be confirmed when all numeric columns are correctly recognized and their values are displayed properly in the preview table. Note that Cactus uses a dot as the decimal separator. If a column you expected to be numeric is imported as text, check the source file for stray characters or an incorrect decimal separator, then re-import the dataset.

5.6 Step 6 - Move to the Edit module

Once the preview looks correct, click **Next step - Edit** to continue.

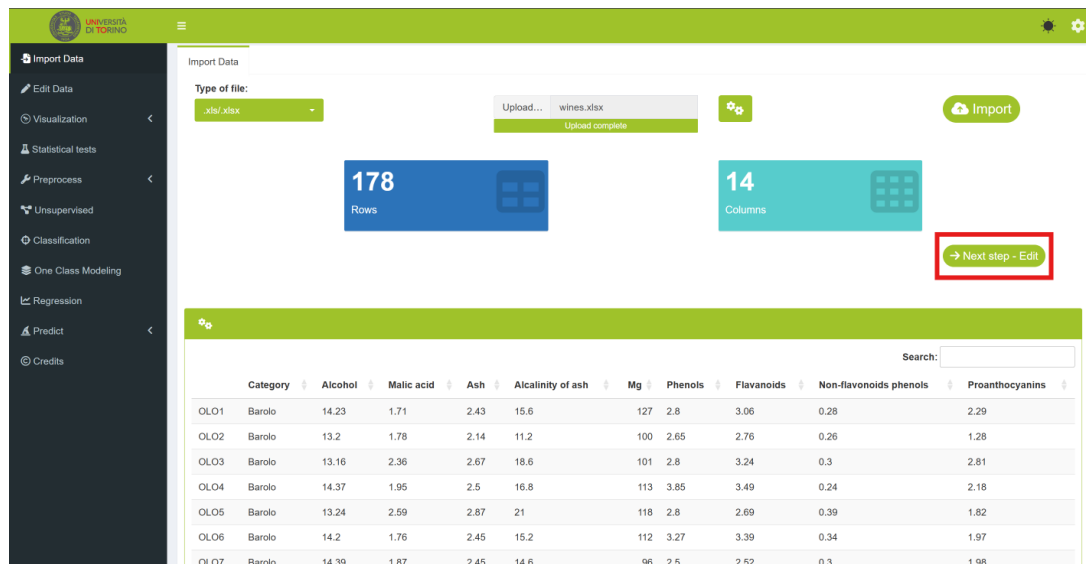


Figure 5.6: Next step - Edit button

5.7 Common mistakes

- Selecting the wrong **Type of file** for the uploaded extension - blocked with a "File type mismatch" pop-up.
- Leaving **Use first column as row labels** checked when that column contains duplicate values - import is blocked with an error asking you to uncheck the option.
- Enabling **Show entire dataset** in the preview settings on a very large dataset - you will be asked to confirm, since this can slow down the browser.
- Forgetting to adjust the **Delimiter/Decimal symbol** for files exported from software using European conventions (semicolon delimiter, comma decimal).

Chapter 6

Edit

The Edit module (sidebar entry **Edit Data**) cleans your imported data and prepares it for analysis. It contains four tabs, which must be completed in order: **Edit**, **Split Train/Test**, **Missing Values**, and **Edited Data**. Each tab becomes available only after the previous step has been completed. The six summary panels shown inside the green box in Figure 6.1 are updated automatically after each subsequent step.

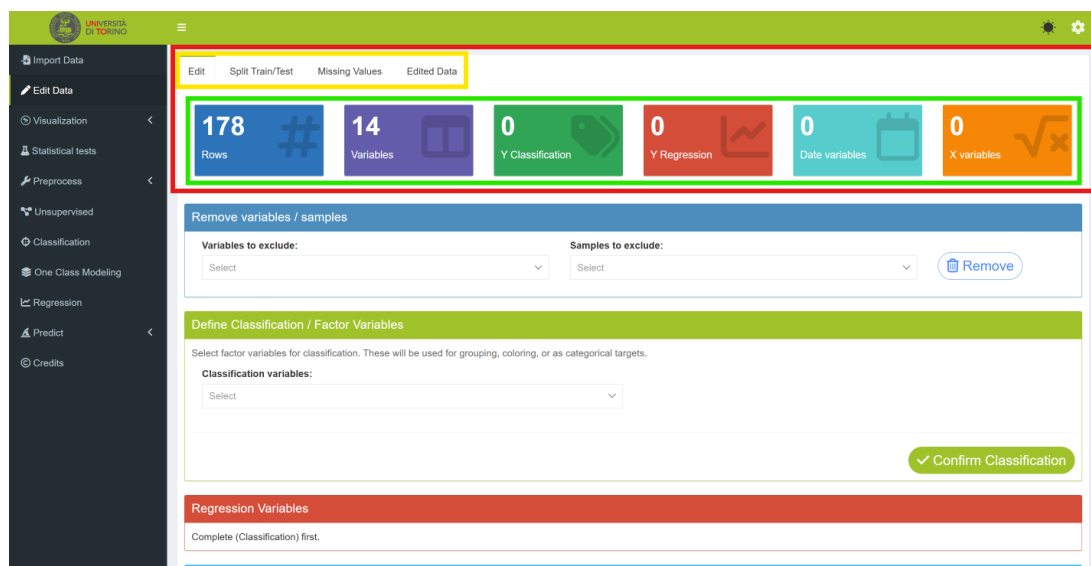


Figure 6.1: Edit Data Steps

6.1 Step 1 - Remove variables or samples (optional)

If some columns or rows should not be included in the analysis, select them and click **Remove**. This step is entirely optional: if there are no variables or samples to remove, proceed without making any changes.

The screenshot shows a panel titled "Remove variables / samples". It contains two dropdown menus: "Variables to exclude:" and "Samples to exclude:", both currently set to "Select". To the right of these is a blue "Remove" button with a trash icon. Below the dropdowns is a yellow box labeled "Removed items:". Inside this box, there are two sections: "Variables:" which lists "Alcohol" and "Malic acid" with red dots next to them, and "Rows:" which lists "OL01" and "OL02" with red dots next to them.

Figure 6.2: Remove variables / samples step

6.2 Step 2 - Confirm Classification variables

Select any columns that contain categorical information (e.g. class labels), choose which groups to keep, and then click **Confirm Classification**. If your dataset does not contain any categorical variables, simply click **Confirm Classification** without selecting any columns.

The screenshot shows a panel titled "Define Classification / Factor Variables". It includes a subtitle: "Select factor variables for classification. These will be used for grouping, coloring, or as categorical targets." There are two main sections: "Classification variables:" with a dropdown menu showing "Category" and a close button "x"; and "Levels for Category:" with a yellow-bordered box containing three items: "Barberis" (selected with a yellow circle), "Barolo", and "Grignolino", each with a close button "x". To the right is a bar chart titled "Distribution for Category:" showing three bars for "Barberis" (orange), "Barolo" (purple), and "Grignolino" (blue). At the bottom right, there is a green button with a checkmark and the text "Confirm Classification", which is highlighted with a red rectangle.

Figure 6.3: Confirm Classification step

Tip

You can also remove all samples belonging to an entire class by clicking the **x** next to that class in the section highlighted in Figure 6.3.

6.3 Step 3 - Confirm Regression variables

From the dropdown menu, select each numeric column that you may want to use as a response variable (Y) in subsequent regression analyses, or as a continuous numeric variable for colouring plots. Then click **Confirm Regression**. If your dataset does not contain any variables of this type, simply click **Confirm Regression** without selecting any columns.

The screenshot shows a panel titled "Define Regression / Numeric Variables". It includes a subtitle: "Select numeric variables for regression analysis or continuous targets." There are two main sections: "Regression variables:" with a dropdown menu showing "Proline" and a close button "x"; and "Range for Proline:" with a yellow-bordered box containing a range slider. The slider has values "278" and "1536.9387755102" at its ends, with intermediate values "478.285714285714" and "1536.9387755102" displayed. To the right is a histogram titled "Distribution for Proline:" showing a distribution of values. At the bottom right, there is a red button with a checkmark and the text "Confirm Regression", which is highlighted with a red rectangle.

Figure 6.4: Confirm Regression step

Tip

If you want to restrict the dataset to a predefined range of response values, use the slider to set the acceptable range for each selected response variable (y). Samples with values below the lower limit or above the upper limit will be removed from the subsequent analyses. This step is optional; leave the full range selected if you do not want to exclude any samples. Figure 6.4.

6.4 Step 4 - Confirm Date variables

If your data contains date/time columns, select them and define a valid range. If no date/time columns are present, click **Skip - Continue**.

Note

Cactus recognises a date/time column if it matches one of the following formats:

- **Dates:** YYYY-MM-DD, DD/MM/YYYY, MM/DD/YYYY, YYYY/MM/DD, DD-MM-YYYY, or MM-DD-YYYY.
- **Dates with time:** any of the date formats above followed by HH:MM or HH:MM:SS (e.g. 2024-03-15 14:30:00).
- **Time only:** HH:MM or HH:MM:SS.
- **Excel date numbers** and **Unix timestamps** expressed in seconds or milliseconds are also detected automatically, as are columns already stored using a proper date/time type.

A column is recognised only if more than half of its values match one of these formats. If a date column is not detected, check that its values are consistent and that different formats are not mixed within the same column, for example 15/03/2024 and 2024-03-15.

Define Date/Time Variables

Select one or more date/time columns. You can define multiple ranges for each.

Date/Time variables:

Data

Range: 1995-05-11 - 1995-05-28

Available: 1995-01-07 to 1995-07-03

Confirm Dates

Define X Variables

Complete (Date variables) first.

SU	MO	TU	WE	TH	FR	SA
30	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31	1	2	3

Figure 6.5: Date variables step

Tip

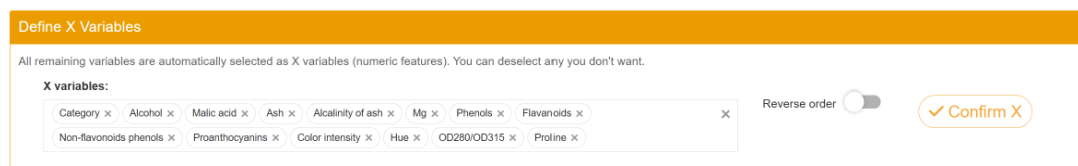
To remove samples according to their date, click and drag along the date-range bar to select the interval that you want to retain. You can define multiple separate date ranges by clicking the + button highlighted in yellow. Use the - button to remove a previously added range. Samples whose dates fall outside the selected ranges will be excluded from the subsequent analyses. Figure 6.5.

6.5 Step 5 - Confirm X (predictor) variables

Review which columns will be used as predictors. By default, all remaining columns are selected. Once the predictor variables have been checked, click **Confirm X**.

Tip

The **Reverse order** switch is particularly useful for spectral plots, as it allows the spectrum to be displayed using the conventional orientation for the selected spectral axis. For example, spectra expressed in wavenumbers (cm^{-1}) are often displayed from higher to lower wavenumber values.



Define X Variables

All remaining variables are automatically selected as X variables (numeric features). You can deselect any you don't want.

X variables:

Category x Alcohol x Malic acid x Ash x Alkalinity of ash x Mg x Phenols x Flavanoids x

Non-flavanoids phenols x Proanthocyanins x Color intensity x Hue x OD280/OD315 x Proline x

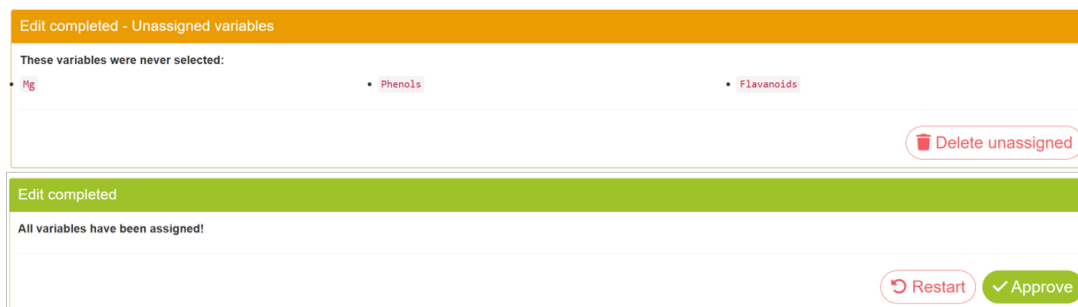
Reverse order ☐

Confirm X

Figure 6.6: Confirm X (predictor variables) step

6.6 Step 6 - Approve

If all variables have been assigned correctly, the message **All variables have been assigned!** will appear. If one or more variables remain unassigned, Cactus will ask whether you want to delete them. You can always edit the configuration again later by clicking **Restart** and repeating the editing process from the beginning. Once everything is correct, click **Approve** to confirm your choices and proceed to the **Split Train/Test** tab.



Edit completed - Unassigned variables

These variables were never selected:

- Mg
- Phenols
- Flavanoids

Delete unassigned

Edit completed

All variables have been assigned!

Restart Approve

Figure 6.7: Approve pipeline step

6.7 Step 7 - Split into Train and Test (optional)

First, decide whether you want to split the dataset into separate **Training** and **Test** sets. If you plan to build and evaluate a predictive model, or simply want to divide your data, click the green **Yes, Enable split** button. To continue using the entire dataset without creating a separate test set, select **No, keep full dataset**, and continue in the Missing Values section.

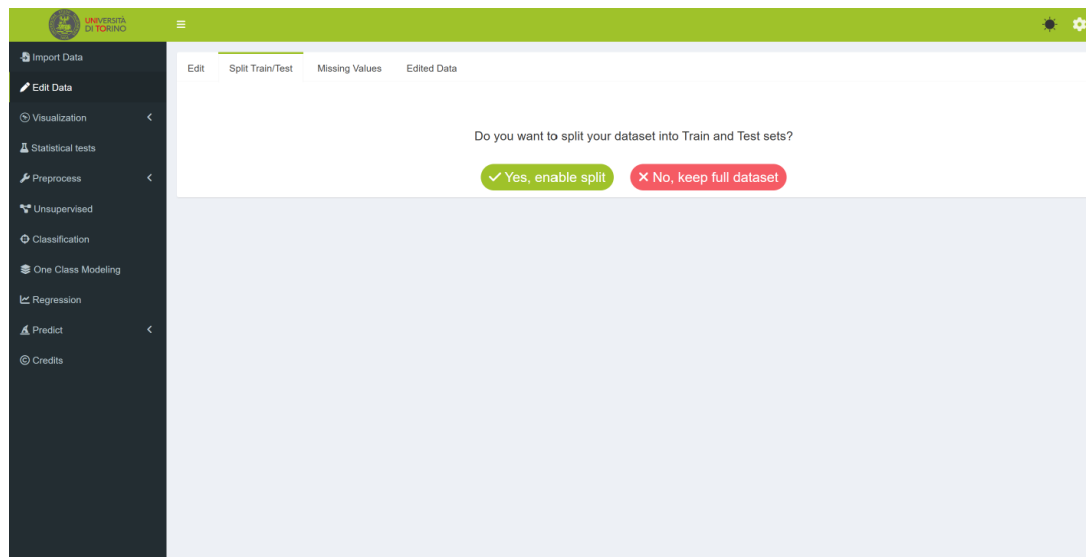


Figure 6.8: Split Train/Test configuration

If splitting is enabled, the **Method** dropdown menu will display the available splitting methods. After selecting a method, use the **Settings** section to configure its specific parameters and define the desired training-set percentage. The information box provides a brief description of the selected method. For a complete explanation of its principles, advantages, and limitations, refer to the bibliography reported in the same box. Once the configuration is complete, click **Run split**.

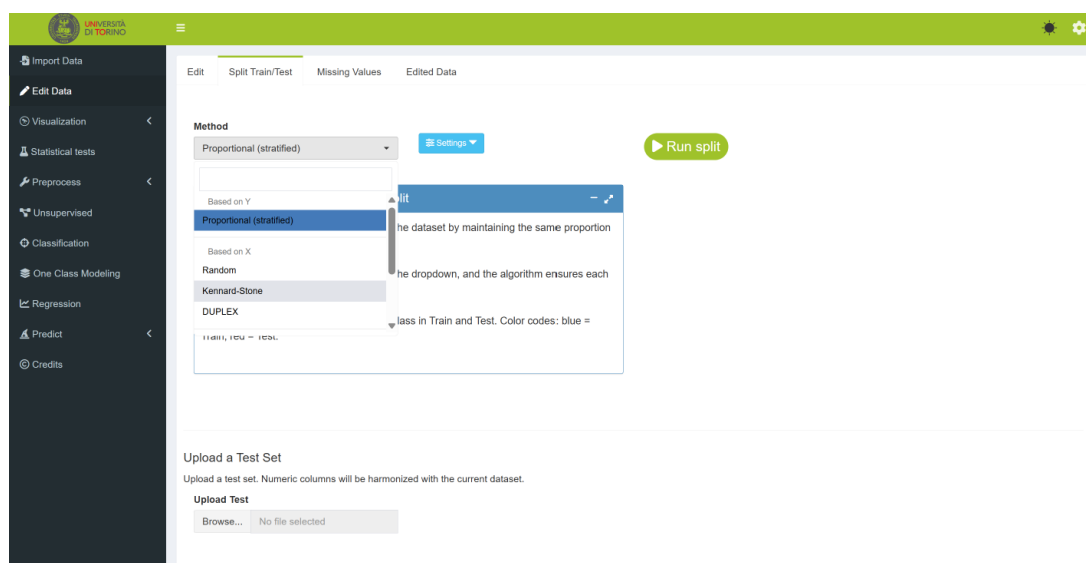


Figure 6.9: Split Train/Test Page

Note

Cactus offers six split methods, grouped by what they use to decide the split:

- **Proportional (stratified)** – based on your Y variable.
- **Random** – based on X, fully
- **Kennard-Stone** – based on X, picks a representative Training set.
- **DUPLEX** – based on X, picks a representative Test set.

- **Sample Order/Time** – based on a date column or import order.
- **Manual** – you choose the Te

Every method except Manual and Sample Order/Time us the same split can be reproduced later.

6.7.1 Proportional (stratified)

This method performs the split according to a selected response variable (Y), preserving its class or value distribution as closely as possible between the **Training** and **Test** sets. It is particularly useful when some classes contain relatively few samples, as it reduces the risk of assigning most or all samples from a rare class to only one of the two sets.

Use the slider to select the percentage of samples assigned to the Training set; the remaining samples will be assigned to the Test set. The default Training percentage is (70%). The **Seed** controls the random component of the split and ensures reproducibility: using the same dataset, settings, and seed value will produce the same Training/Test split.

The method can also be applied to a continuous numeric (Y) variable. In this case, its values are automatically divided into a predefined number of bins (**Y bins**, default: 10), and the split is then stratified according to these intervals.

The optional **Grouping factor (Replicates)** can be used to keep related observations together. Replicate information must already be included as a column in the original imported dataset. All measurements belonging to the same sample or replicate group must have exactly the same label, as shown in Figure 6.10b. Cactus will then assign all rows sharing that label entirely to either the Training or Test set, preventing repeated measurements of the same sample from being divided between the two sets.

(a) Settings available for the Proportional (stratified) splitting method.

ID	Category	Replicates
OLO1	Barolo	Wine1
OLO2	Barolo	Wine1
OLO3	Barolo	Wine1
OLO4	Barolo	Wine1
OLO5	Barolo	Wine1
OLO6	Barolo	Wine2
OLO7	Barolo	Wine2
OLO8	Barolo	Wine2
OLO9	Barolo	Wine2
OLO10	Barolo	Wine2

(b) Example of replicate labels included in the original dataset. Rows belonging to the same replicate group must share the same label.

Figure 6.10: Proportional stratification settings and example of replicate-group labels.

6.7.2 Random

Splits your data completely at random, according to the **Train percentage**. The **Seed** controls the random component of the split and ensures reproducibility: using the same dataset, settings, and seed value will produce the same Training/Test split. An optional **Grouping factor (Replicates)** keeps them, exactly as in Proportional.

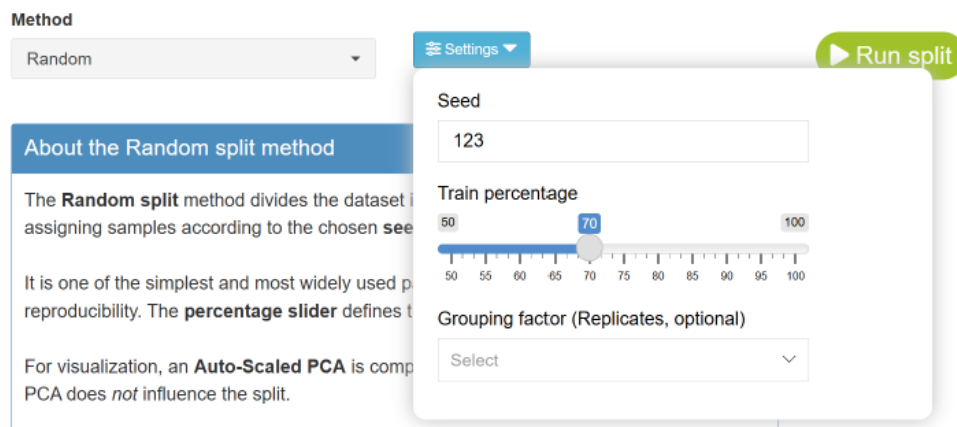


Figure 6.11: Random split settings

6.7.3 Kennard–Stone

Unlike random splitting methods, Kennard–Stone selects samples according to their position in the predictor space (X). The algorithm builds a Training set whose samples are distributed as evenly as possible across the available data space. This generally produces a representative Training set and reduces the risk of obtaining an unbalanced split due to random selection.

The available settings are:

- **Distance metric** - Defines how the distance between samples is calculated, Mahalanobis or Eclidean.
- **Preprocessing** - Applies **None**, **Center**, or **Autoscale** to the predictor variables before calculating the distances.
- **Use PCA** - Performs sample selection in PCA score space instead of using the original predictor variables. This option is enabled by default and can be useful when the dataset contains many variables or highly correlated predictors. The number of components can be specified directly or as a fraction of explained variance, for example 0.95 to retain 95% of the variance.
- **Grouping factor (Replicates)** - same that Proportional case.
- **Initial samples** - Optionally specifies samples that must be included in the Training set before the automatic selection begins.

Method

Kennard-Stone

About the Kennard-Stone split

The **Kennard-Stone algorithm** selects training samples uniformly as possible. It iteratively chooses samples using Euclidean or Mahalanobis distance.

This deterministic method ensures the training samples are representative while remaining samples form the test set.

Reference: Kennard, R. W., & Stone, L. A. (1968). [Experiments.](#)

Upload a Test Set

Upload a test set. Numeric columns will be harmonized.

Upload Test

Browse... No file selected

Settings

Seed

123

Train percentage

50 70 100

Distance metric

☒ Mahalanobis ☐ Euclidean

Preprocessing

☐ None ☒ Center ☐ Autoscale

☒ Use PCA

PC components (count or <1 fraction)

0,95

Selected: 1 PCs (approx 95.4% cumulative variance)

Grouping factor (Replicates)

Select

Initial samples (optional, ordered)

Select

Run split

Figure 6.12: Kennard-Stone split settings

Warning

Kennard–Stone cannot be used when (X) contains missing values. Handle them first, save the imputed dataset, and then restart the app and return to the split step. Be careful: imputing before splitting may introduce **data leakage**.

6.7.4 DUPLEX

DUPLEX is a variant of Kennard-Stone that distributes both the Training and Test samples evenly across the predictor space (X). Unlike Kennard–Stone, it also aims to make the Test set representative rather than simply assigning the remaining samples to it.

It uses the same settings as Kennard-Stone, including distance metric, preprocessing, PCA, grouping factor, and initial samples. The same restriction regarding missing values also applies.

Method

DUPLEX

About the DUPLEX algorithm

The **DUPLEX** method extends Kennard-Stone to create training and test sets that are both representative of the data.

It ensures diversity in both subsets and avoids correlation in the partition.

Reference: Snee, R. D. (1977). *Validation of regression models*.

Settings

Seed: 123

Train percentage: 70%

Distance metric: ☒ Mahalanobis ☐ Euclidean

Preprocessing: ☐ None ☒ Center ☐ Autoscale

☒ Use PCA

PC components (count or <1 fraction): 0.95

Selected: 1 PCs (approx 95.4% cumulative variance)

Grouping factor (Replicates): Select

Run split

Figure 6.13: DUPLEX split settings

6.7.5 Sample Order/Time

Splits the dataset according to a selected date/time variable or, if none is selected, according to the original sample order. The first percentage of samples is assigned to the Training set, while the remaining samples are assigned to the Test set.

This method does not use a random seed.

Method

Sample Order/Time

About the Sample Order/Time split

The **Sample Order/Time split** divides the dataset according to a selected **date/time variable**. The first $p\%$ of samples go to **Test**.

This approach preserves the original sequence of samples to predict earlier ones, reducing the risk of data leakage.

If a **date/time variable** is selected, the dataset is sorted by that column before splitting. Otherwise, the current **sample order** is preserved.

The visualization shows the sample sequence on the X-axis and the corresponding ordered rank on the Y-axis. A dashed vertical line marks the Train/Test boundary.

Settings

Train percentage: 70%

Order variable: Sample order

The dataset is sorted by this variable and split at the chosen proportion. No randomness involved.

Run split

Figure 6.14: Sample Order/Time split settings

6.7.6 Manual

Allows you to search for and manually select the rows assigned to the Test set. All remaining rows are assigned to the Training set.

This method is useful when you already know which samples should be held out.

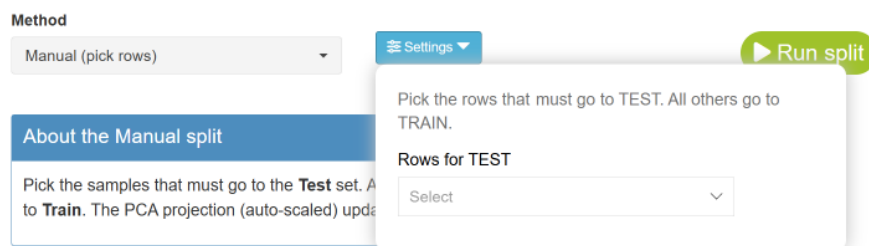


Figure 6.15: Manual split row selector

6.8 Run Split and Graphical Overview

After configuring the selected splitting method, click **Run Split** to generate the Training and Test sets.

Cactus then displays a graphical summary of the split. Depending on the selected method, the plot shows either the proportions of samples assigned to the Training and Test sets or a scatter plot illustrating how the samples were selected. Training samples are shown in **blue**, while Test samples are shown in **red**.

The visualization can be used to verify that the split is consistent with the chosen settings before continuing with the analysis.

Tip

If you entered the Split module by mistake and do not want to create a Test set, select either the **Random** or **Proportional** method and set the Training percentage to **100%**. All samples will remain in the Training set, allowing you to continue as if you had selected **No, keep full dataset** before entering the module.

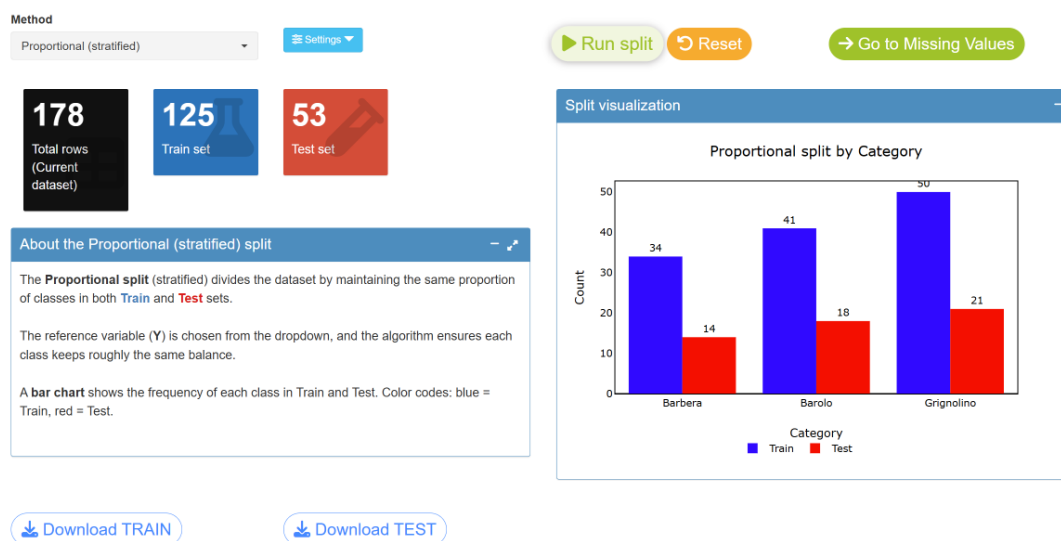


Figure 6.16: Split results and graphical overview

6.8.1 Uploading a Test Set

Instead of, or in addition to, splitting the main dataset, you can upload a separate file to use as an uploaded Test set from the **Upload a Test Set** tab.

The same file formats supported by the Import module are accepted (.csv, .txt, .xls, and .xlsx), together with the same advanced options, including delimiter, decimal symbol, row labels, header, and empty-value settings.

Before loading the file, Cactus compares it with the main dataset and performs the following checks:

- **Missing columns** - If the uploaded file does not contain a column present in the main dataset, the file is rejected and Cactus lists the missing columns.
- **Extra columns** - Columns present in the uploaded file but not in the main dataset are automatically removed, and a warning is displayed.
- **Type mismatches** - If a column has a different data type from the corresponding column in the main dataset, Cactus attempts to convert it. Values that cannot be converted are replaced with missing values, and a warning reports how many values were affected.
- **Duplicate row names** - If a sample name already exists in the main dataset, it is automatically modified to avoid duplicated row names.

Click **Load Test Set** to confirm the upload. A summary reports the number of loaded samples and columns and displays any warnings related to automatic corrections. Click **Delete Test Set** to remove the uploaded Test set.



Figure 6.17: Upload a Test Set panel

Tip

An uploaded Test set is stored separately from the internal Train/Test split. To use only the uploaded Test set, set the internal Training percentage to **100%**. Otherwise, the main dataset will also be split.

6.9 Step 8 - Handle missing values

Once your data is split, Cactus scans it for missing values (NAs). If none are found, you will see a green "No missing values detected" message – just click **Continue to Edited Data** and skip the rest of this step.

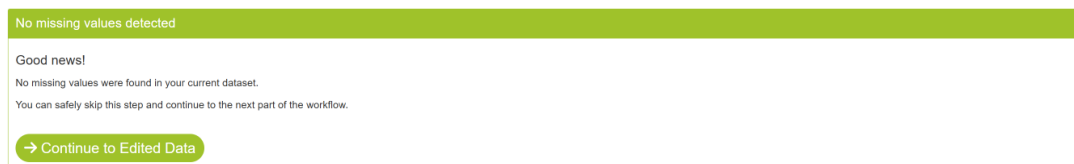


Figure 6.18: No missing values detected

Warning

Imputed values are **temporary** and only affect Visualization, Statistics and Unsupervised exploration. When you later train a supervised model (Classification, One Class Modeling, Regression) with cross-validation, your chosen imputation method is re-applied independently within each fold, so it never "leaks" information from one fold into another.

6.9.1 Missing values overview

At the top of the tab, a summary line shows the total number of unresolved missing values and how many columns they affect. Below it, two tables help you decide what to do:

- **Summary table** - one row per column, with the count and percentage of missing values overall, and separately for Train and Test (if a split exists). Cells are colour-coded: white = 0%, green < 30%, gold 30–50%, red > 50%. If a column is mostly missing, it is usually better to go back to the **Edit** tab and delete it instead of imputing it.
- **Positions table** - lists every individual missing cell (row name, column, type), and, once imputed, the value that was filled in and which method was used. Cells that have already been imputed are highlighted in gold.

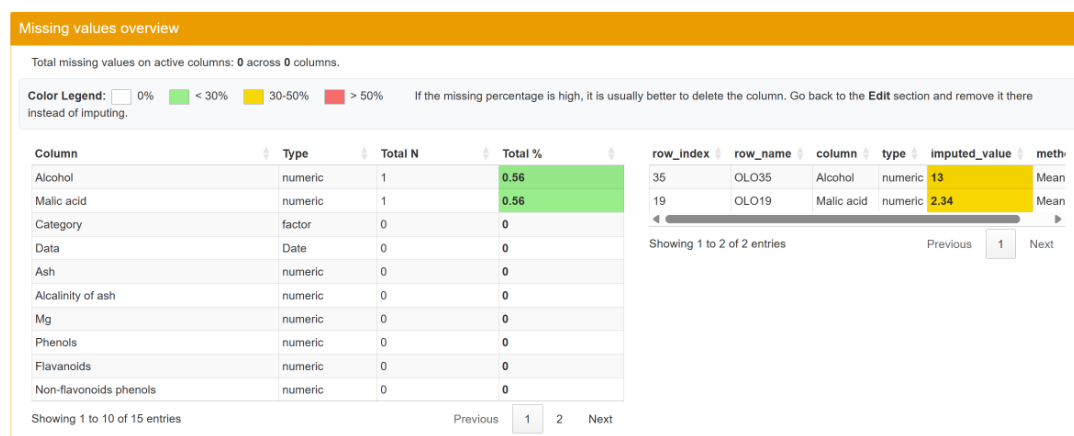


Figure 6.19: Missing values overview: summary and positions tables

6.9.2 Choosing an imputation method

Cactus offers six imputation methods, selected from the **Imputation method** dropdown:

Note

- **Mean (numeric)** – fills gaps with the column average.
- **Median (numeric)** – fills gaps with the column median (more robust to outliers).
- **KNN (numeric)** – estimates each value from similar samples.
- **Random (LOD–LOQ range)** – fills gaps with a random value in a chosen range.
- **Interpolation (spectral/adjacent)** – fills gaps using neighbouring columns.
- **Manual (single cell override)** – you type the replacement value yourself.

For every method except Manual, select the **Numeric columns to impute** and click **Run imputation**; a description of the selected method is also shown live on the tab.

Figure 6.20: Imputation method dropdown and column picker

6.9.3 Mean

Replaces missing values in the selected numeric columns with the column’s average, calculated only from the **Train** rows. The same value is then used for both Train and Test rows.

6.9.4 Median

Same idea as Mean, but uses the median instead of the average - calculated on Train rows only and applied to both Train and Test.

6.9.5 KNN

Estimates each missing value from the **K** most similar samples (nearest neighbours), based on the other numeric columns. Set **Number of neighbors (K)** (default 5). The neighbour search is fitted on Train rows and then applied to all rows.

Warning

KNN cannot use your classification, regression (Y), or date columns as imputation targets - Cactus removes them automatically from the selection and warns you if you tried to include one.

6.9.6 Random (LOD–LOQ range)

Fills gaps with a random value drawn from a uniform distribution between a **Random min value** and **Random max value** that you set – typically the instrument’s limit of detection/quantification (LOD–LOQ). A **Random seed** keeps the result reproducible.

Imputation missing values

Imputation method: Random (LOD-LOQ range)

Numeric columns to impute: Alcohol, Malic acid

Random min value: 0

Random max value: 1

Random seed: 123

[Run imputation](#)

Method description

Random imputation (LOD-LOQ range)
 Missing values in the selected numeric columns are replaced by random values drawn from a uniform distribution between the user-specified **min** and **max** (typically the LOD-LOQ range), using a reproducible random seed. *Imputed values will match the decimal precision of existing data.*

Figure 6.21: Random (LOD-LOQ) imputation settings

6.9.7 Interpolation (spectral/adjacent)

Fills a missing value by looking at the nearest non-missing *columns* to its left and right (by column position) and interpolating linearly between them; if only one side has data, that value is used directly. This is the recommended method for **spectroscopic data** (NIR, Raman, UV-Vis), where the signal varies smoothly across wavelengths. It is applied directly, with no training statistics involved, so there is no risk of data leakage during cross-validation.

6.9.8 Manual (single cell override)

Pick a **Column** and a **Row** (by name) that has a missing value, type the **New value**, then click **Save imputation (preview)**. Nothing is applied immediately – it is only saved as a preview, so you can review every manual change before committing them all at once in the next step.

6.10 Step 9 - Confirm and continue

Once every missing value has a method assigned, the **Confirm imputation & proceed** button becomes active (it stays disabled as long as any value is still unresolved). Use **Reset all imputations** at any point to discard everything and start over.

Actions

[Reset all imputations](#) [Confirm imputation & proceed](#)

Figure 6.22: Missing Values overview and imputation controls

6.11 Step 9 - Approve the final dataset

On the **Edited Data** tab, use the radio buttons to switch between the available datasets: **Full**, **Training**, and **Test**. Check the preview table, and use the **Settings** menu, identified by the gear icon, to download the selected dataset. Once everything has been verified, click **Approve Dataset** to unlock the remaining sections of the application.

Dataset: ☒ Full ☐ Train ☐ Test

✓ Approve Dataset

Edited Data

Show number of rows: 20

Show number of columns: 10

☐ Show entire dataset

Download CSV

		Malic acid	Ash	Alkalinity of ash	Mg	Phenols	Flavanoids	Non-flavonoids phenols	Proanthocyanins
		2.36	2.67	18.6	101	2.8	3.24	0.3	2.81
		1.95	2.5	16.8	113	3.85	3.49	0.24	2.18
		2.59	2.87	21	118	2.8	2.69	0.39	1.82
		1.87	2.45	14.6	96	2.5	2.52	0.3	1.98
OLO8	Barolo	14.06	2.15	2.61	17.6	121	2.6	2.51	0.31
OLO9	Barolo	14.83	1.64	2.17	14	97	2.8	2.98	0.29
OLO10	Barolo	13.86	1.35	2.27	16	98	2.98	3.15	0.22
OLO11	Barolo	14.1	2.16	2.3	18	105	2.95	3.32	0.22
OLO12	Barolo	14.12	1.48	2.32	16.8	95	2.2	2.43	0.26
OLO13	Barolo	13.75	1.73	2.41	16	89	2.6	2.76	0.29

Showing 1 to 10 of 20 entries

Previous 1 2 Next

Figure 6.23: Edited Data preview with Approve Dataset button

Tip

Approving the dataset again later (after making more changes) resets any models you have already trained - this keeps your results consistent with the data.

6.12 Common mistakes

- Leaving a column unassigned - Cactus will ask you to delete it before you can Approve.
- Choosing **Kennard-Stone** or **DUPLEX** as a split method while missing values still exist - resolve missing values first, or pick a different method.
- Uploading an external test set that is missing a column from the main dataset - rejected outright.
- Trying to impute classification/date columns with **KNN** - not available for those columns.

Chapter 7

Visualization

The Visualization module lets you look at your data as charts. It has three sidebar sub-items: **One Variable**, **Two Variables** and **Multiple Variables**.

Note

Every chart is interactive: hover for details, zoom, pan, and export it as an image. This works the same way everywhere in Cactus - see Chapter 16 for the full guide.

7.1 Step 1 - Pick a tab

Choose **One Variable**, **Two Variables**, or **Multiple Variables** from the sidebar, depending on how many columns you want to compare at once.

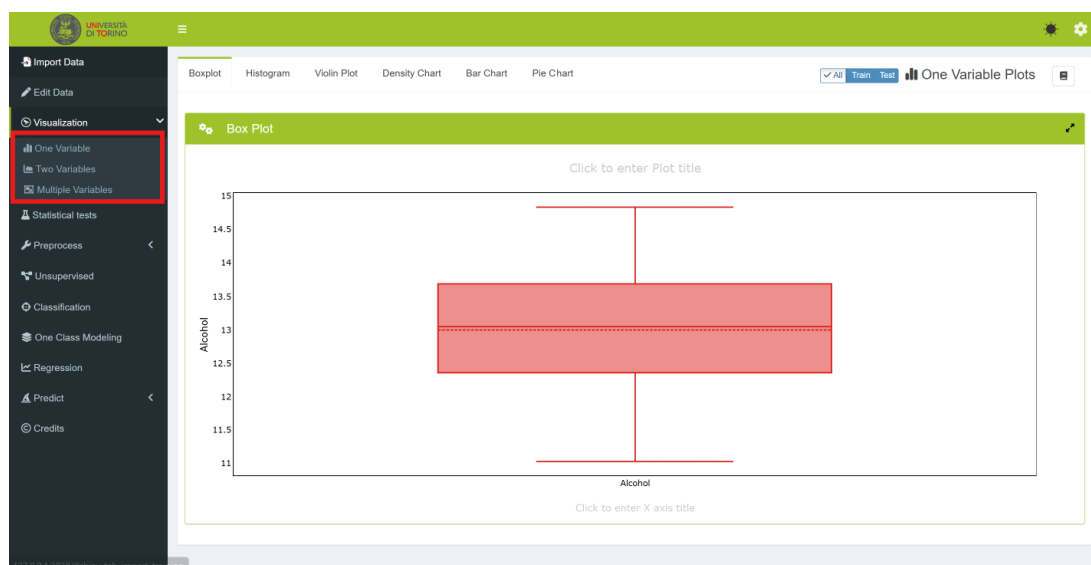


Figure 7.1: Visualization sidebar and tab selection

7.2 Step 2 - Pick a chart type

Each tab has its own set of chart types along the top. Click one to open it.

Tab	Chart types
One Variable	Boxplot, Histogram, Violin Plot, Density Chart, Bar Chart, Pie Chart
Two Variables	Scatter Plot, Bubble Plot, 2D Histogram, 2D Density Chart, Scatter Matrix, Time Series
Multiple Variables	3D Scatter Plot, Boxplots (multi), Heatmap, Spectra / Parallel Coordinates

7.3 Step 3 - Choose your variables or Adjust the chart

Open the **Settings** dropdown (gear-icon) to customize the appearance of the selected chart. Depending on the chart type, you can adjust options such as point size, number of bins, line appearance, transparency, labels, and other display parameters. The chart is updated automatically whenever a setting is changed.

Most charts also allow observations to be coloured according to either a categorical or continuous variable. Categorical variables can be used to distinguish classes or groups, with the option to customise the colour assigned to each category. Continuous numeric variables are instead represented using a colour gradient, making it easier to identify changes, trends, or patterns across their range of values.



Figure 7.2: Chart settings dropdown

7.4 Step 4 - Read the chart

Hover over any point, bar or line to see its exact value. Use the toolbar on the right to zoom, reset, or save the chart as an image (Chapter 16).

7.4.1 One Variable

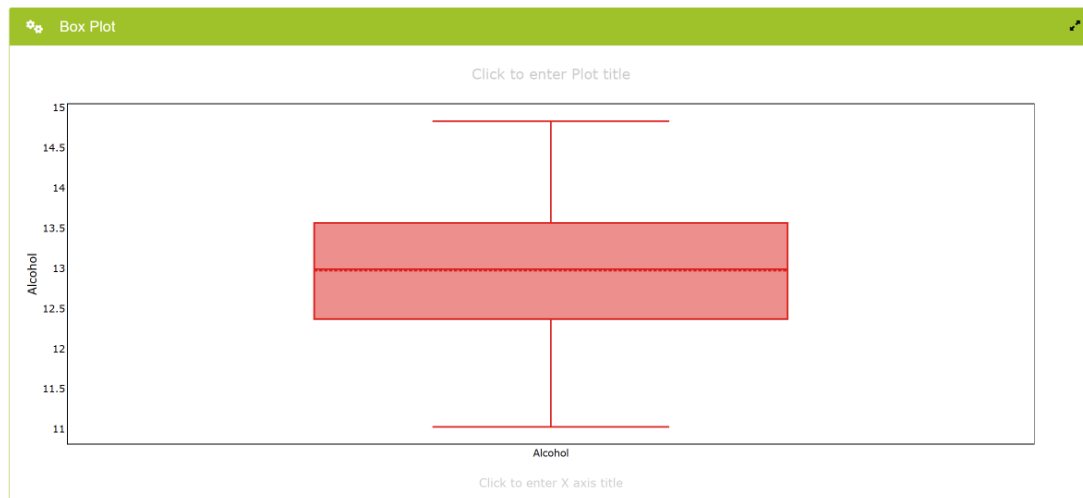


Figure 7.3: Boxplot



Figure 7.4: Histogram

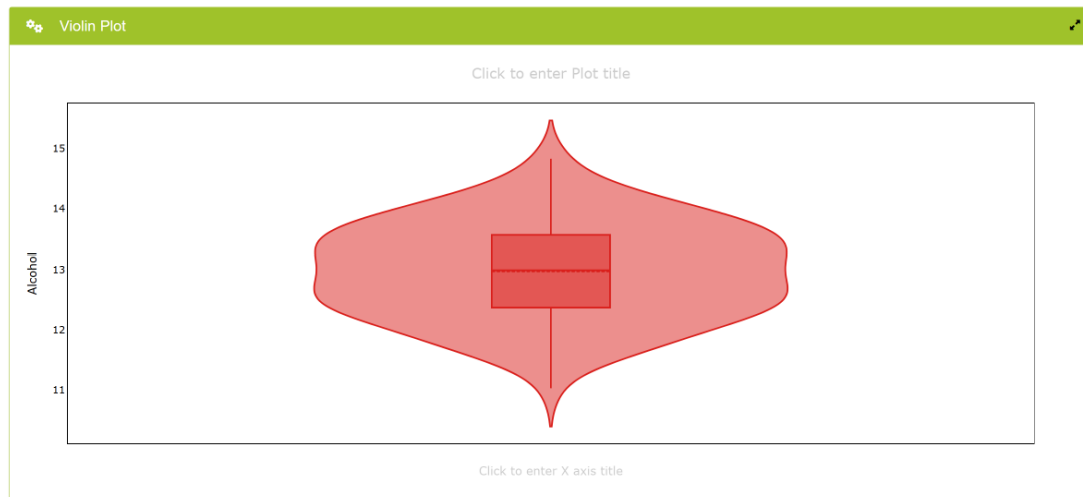


Figure 7.5: Violin Plot

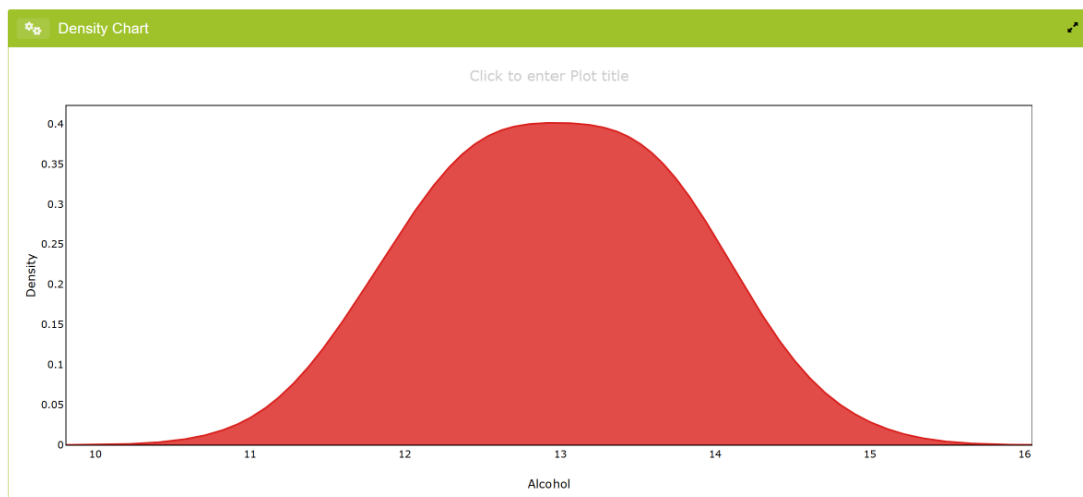


Figure 7.6: Density Chart

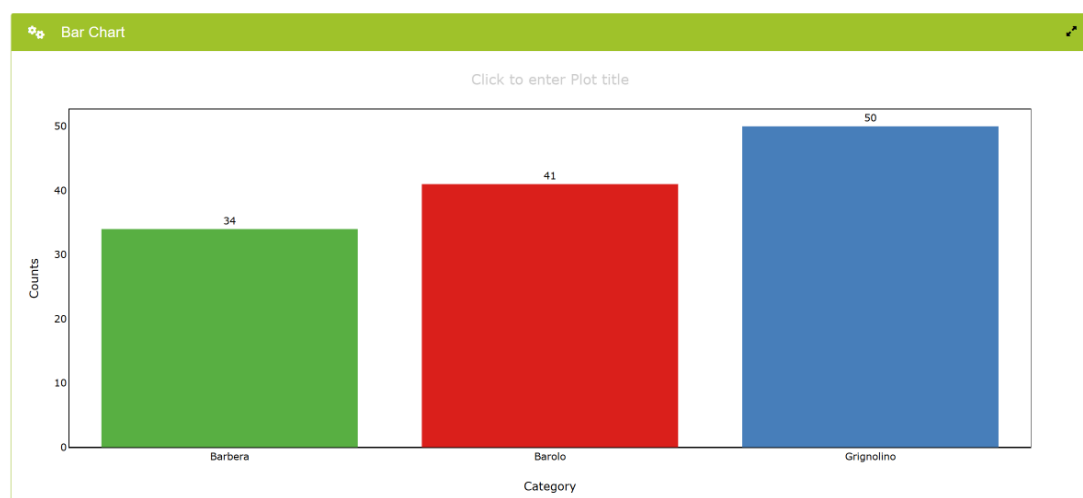


Figure 7.7: Bar Chart

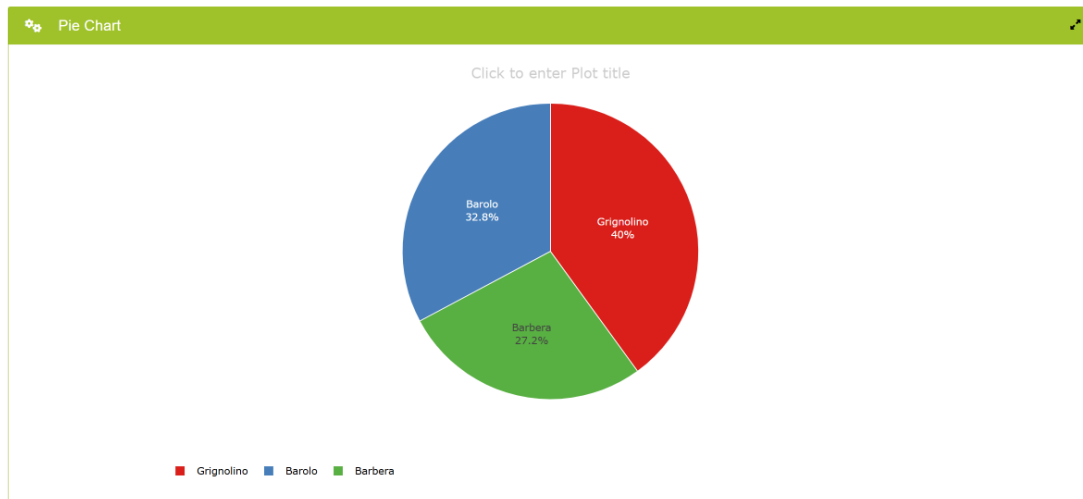
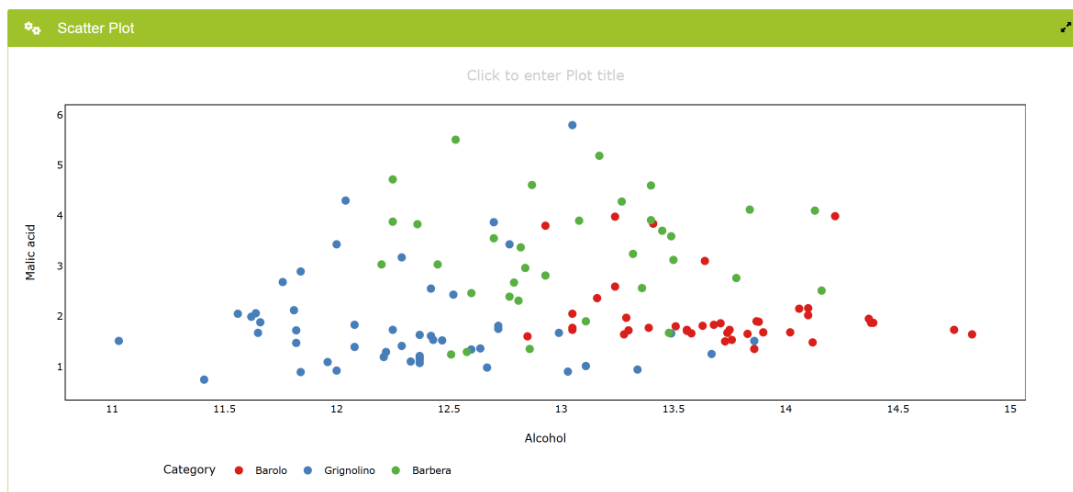


Figure 7.8: Pie Chart

7.4.2 Two Variables



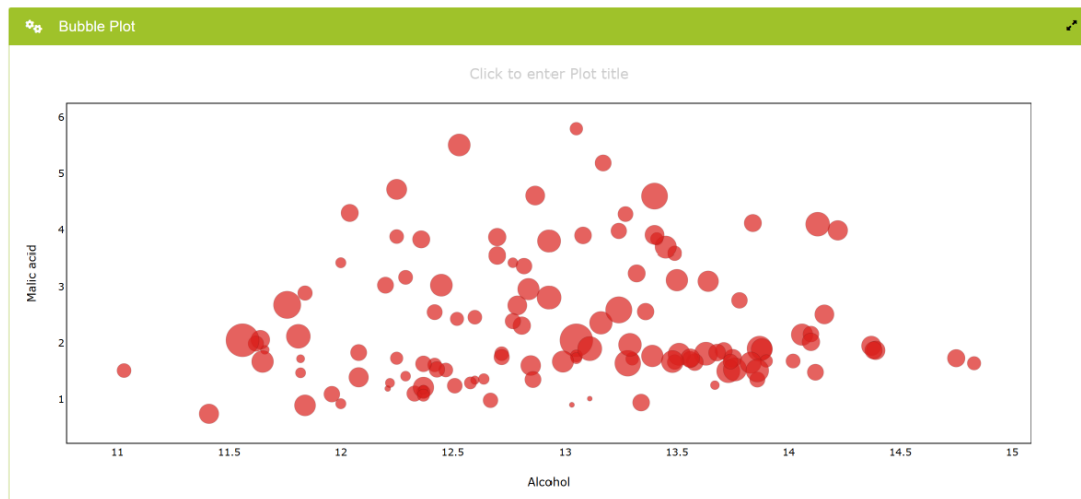


Figure 7.10: Bubble Plot

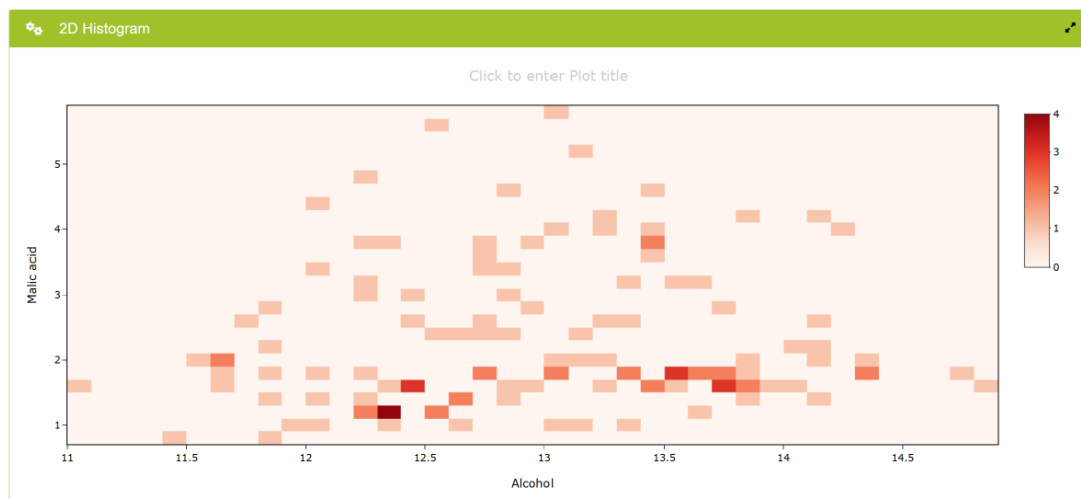


Figure 7.11: 2D Histogram

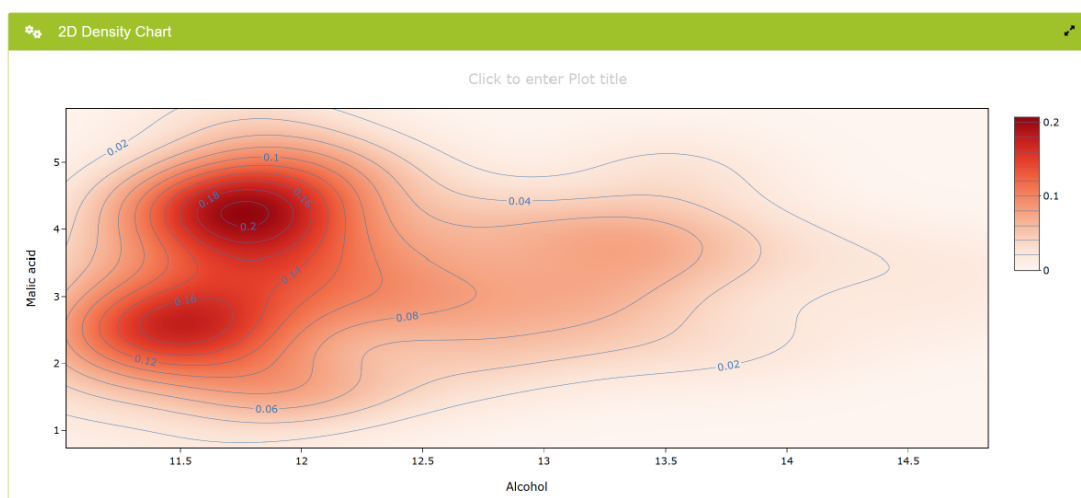


Figure 7.12: 2D Density Chart

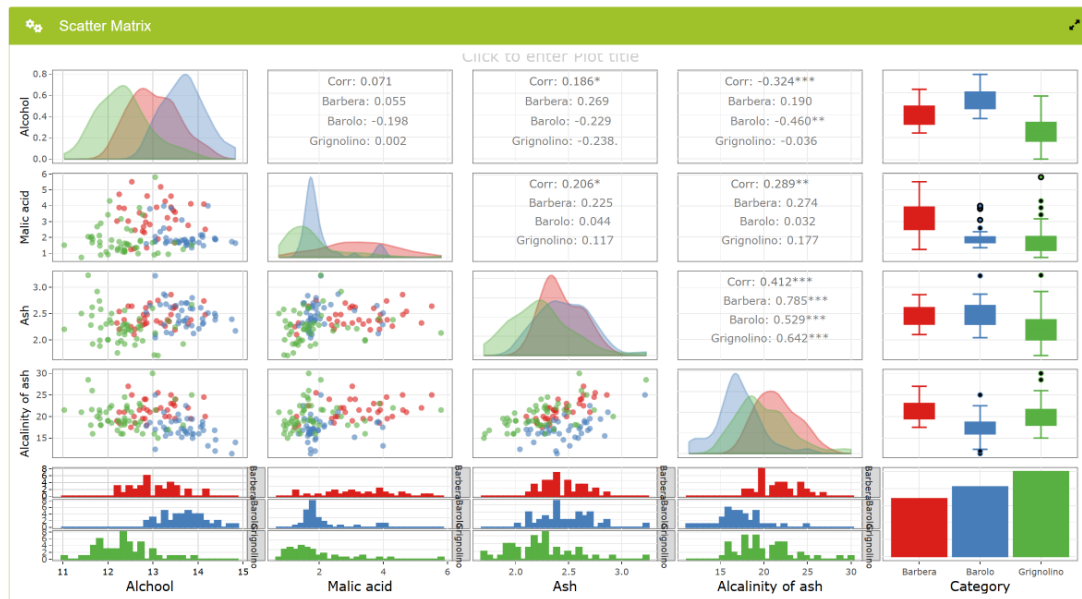


Figure 7.13: Scatter Matrix

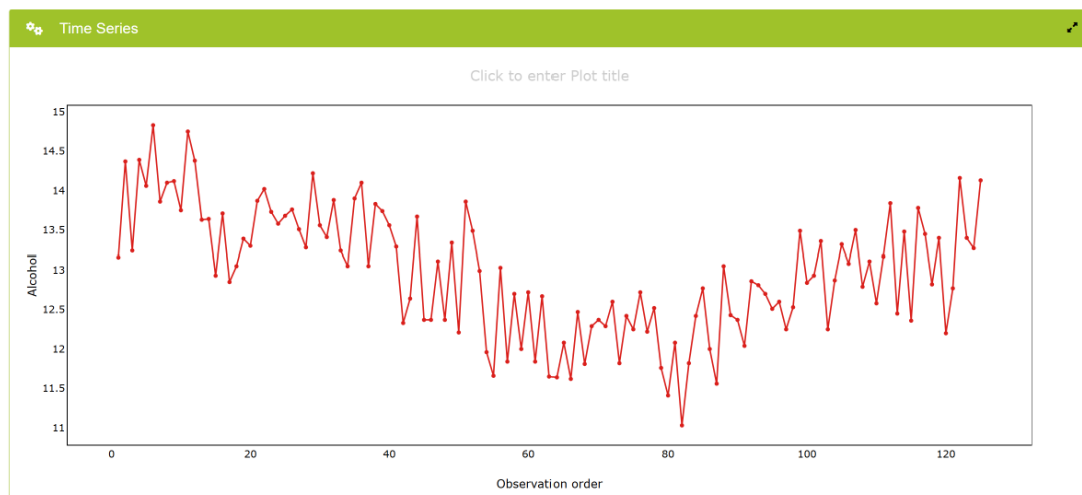


Figure 7.14: Time Series

7.4.3 Multiple Variables

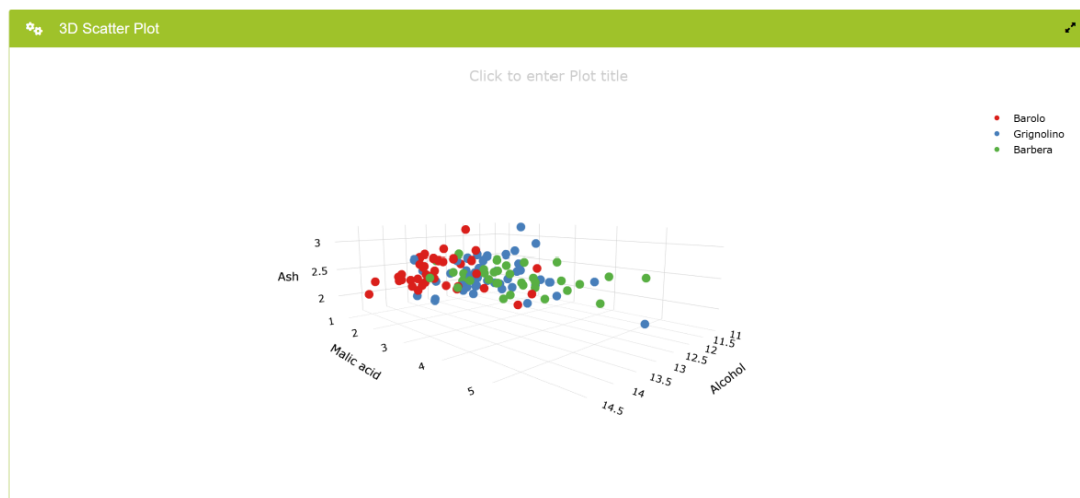


Figure 7.15: 3D Scatter Plot

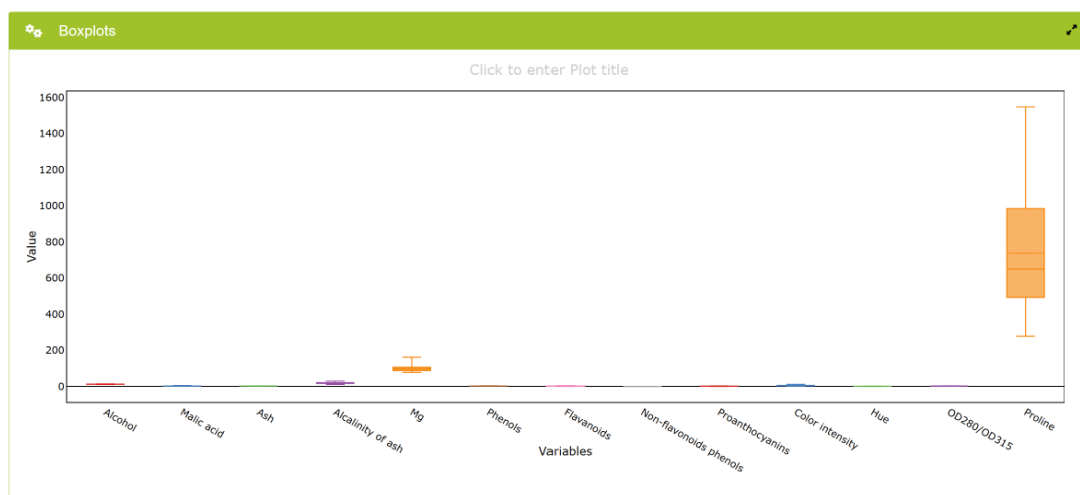


Figure 7.16: Boxplots (multi)

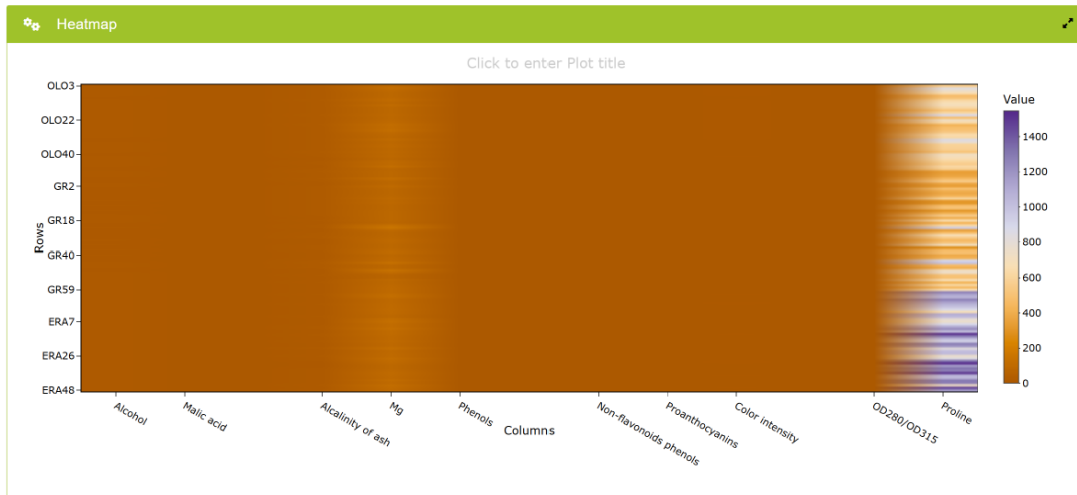


Figure 7.17: Heatmap

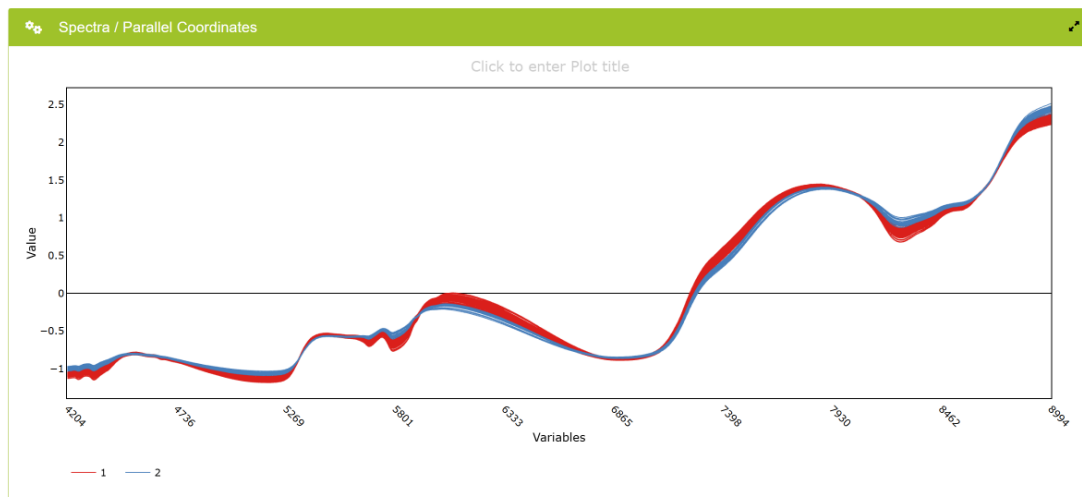


Figure 7.18: Spectra / Parallel Coordinates

7.5 Visualization Module After Train/Test Splitting or Preprocessing

When a Training/Test split has been created or preprocessing has been applied, additional buttons appear in the green bar at the top of the visualization panel. These buttons allow you to switch directly between the available datasets, such as the **Full**, **Training**, **Test**, and **Preprocessed** datasets.

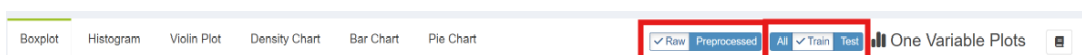
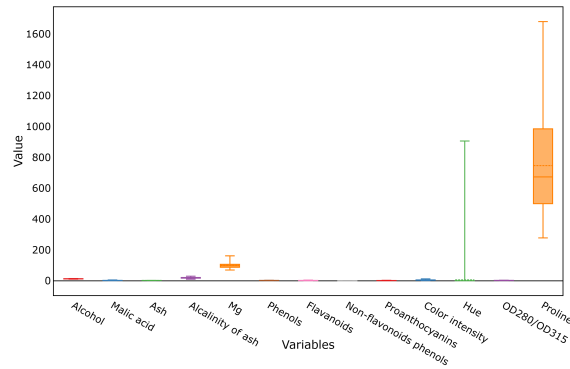
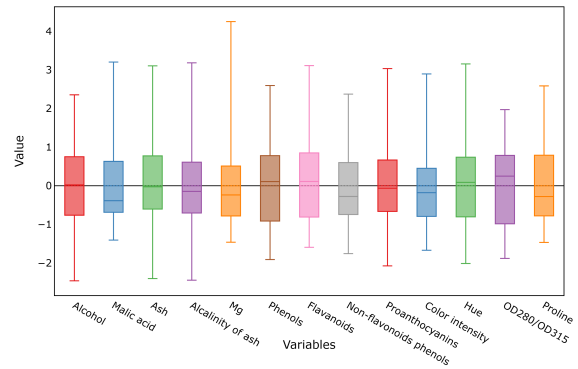


Figure 7.19: Different plot based on preprocessing and splitting

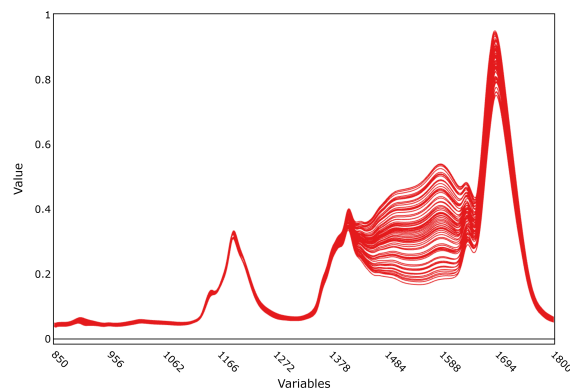
This makes it possible to compare the Training and Test groups visually, check whether the split produced representative subsets, and evaluate how preprocessing has modified the structure, distribution, or overall appearance of the data.



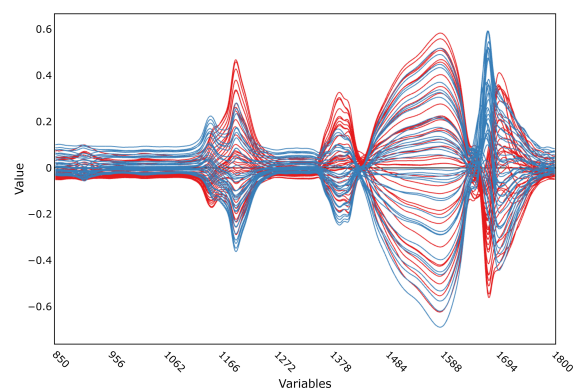
(a) Wines dataset raw data



(b) Wines dataset after autoscaling



(c) NIR Raw spectra



(d) NIR after SNV + Mean Centering

Figure 7.20: Comparison of the available dataset views in the Visualization module.

7.6 Common mistakes

- Comparing groups of very different sizes without checking how many samples are in each.
- Forgetting to click **Plot** again on the Scatter Matrix after changing your variable selection - it does not update automatically.
- Using a Pie Chart for a variable with many categories, which makes slices hard to compare.
- Choosing "Original order" on a Time Series chart and mistaking it for a real time axis - only an actual date column gives a true time axis.

Chapter 8

Statistics Module

The Statistics module (sidebar entry **Statistical tests**) runs standard statistical tests on your data. Unlike Visualization, nothing is calculated until you click a button.

8.1 Step 1 – Pick a test

Choose a tab: **Normality**, **t-test**, **ANOVA**, **Correlation**, or **Volcano Plot**.

Test	What it tells you
Normality	Whether a variable follows a "normal" bell-curve distribution.
t-test	Whether two groups differ, for one variable.
ANOVA	Whether three or more groups differ, for one variable.
Correlation	How strongly pairs of variables (or samples) move together.
Volcano Plot	Which variables differ the most between two groups, screening many at once.

8.2 Step 2 – Choose your variable(s)

Select the numeric variable(s) to test, and a grouping/category column where needed.

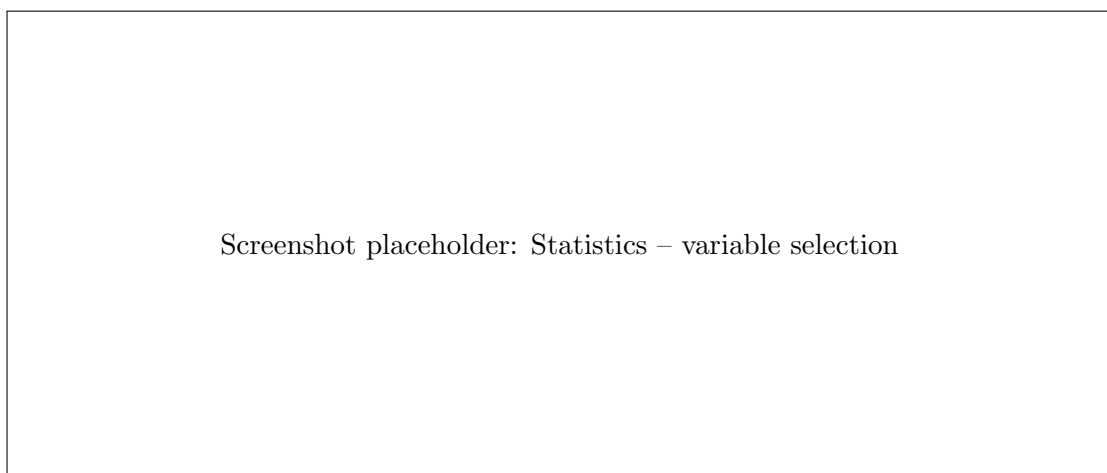


Figure 8.1: Statistics – variable selection interface.

8.3 Step 3 – Run the test

Click **Calculate** (or **Test**). Results appear below: a table with a p-value, and usually a chart.

Tip

A p-value below 0.05 is usually read as "a real difference/effect" (marked with stars: * = $p \leq 0.05$, ** = $p \leq 0.01$, *** = $p \leq 0.001$). A p-value of 0.05 or above means no difference was detected.

8.3.1 Normality

Placeholder for Figure 8.2: Normality tab with Q-Q plot interface.

Figure 8.2: Normality tab with Q-Q plot interface.

8.3.2 t-test

Placeholder for Figure 8.3: t-test tab with box plot interface.

Figure 8.3: t-test tab with box plot interface.

8.3.3 ANOVA

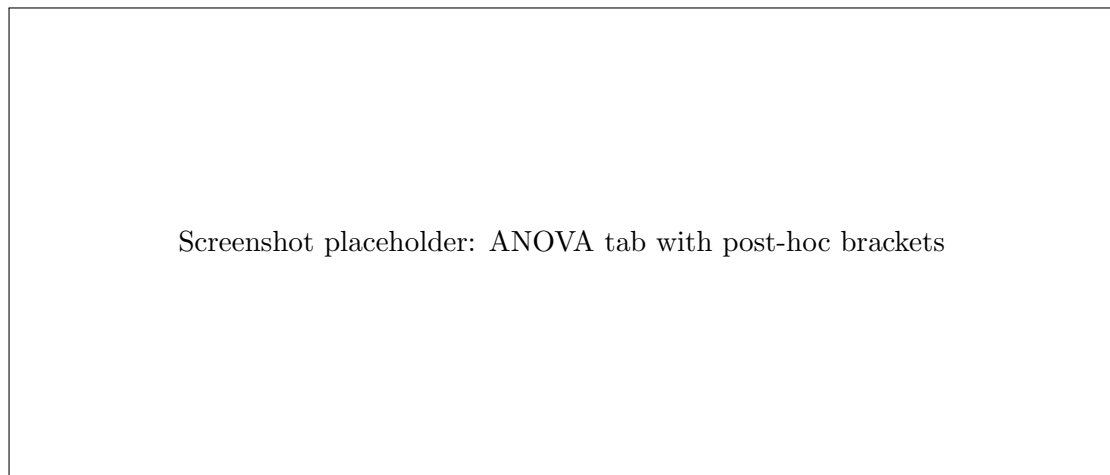


Figure 8.4: ANOVA tab with post-hoc brackets interface.

8.3.4 Correlation

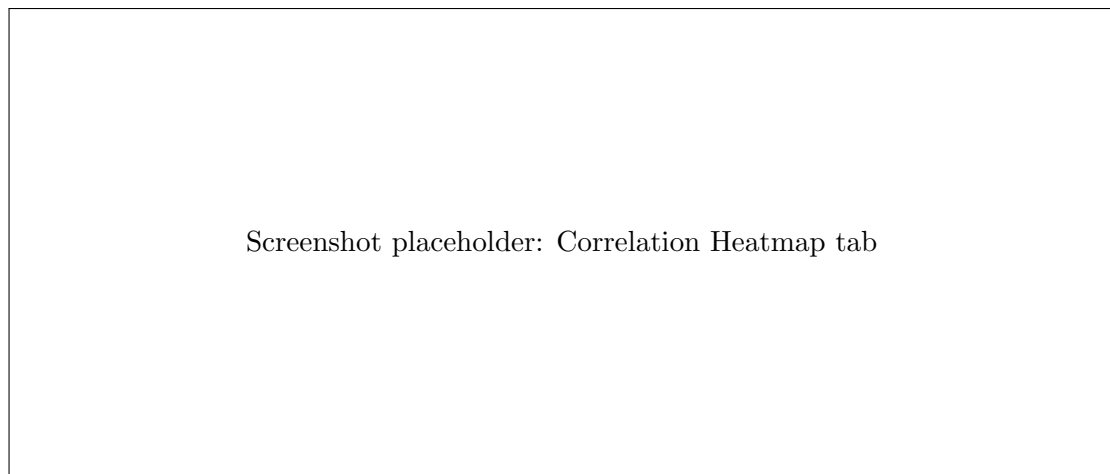


Figure 8.5: Correlation Heatmap tab interface.

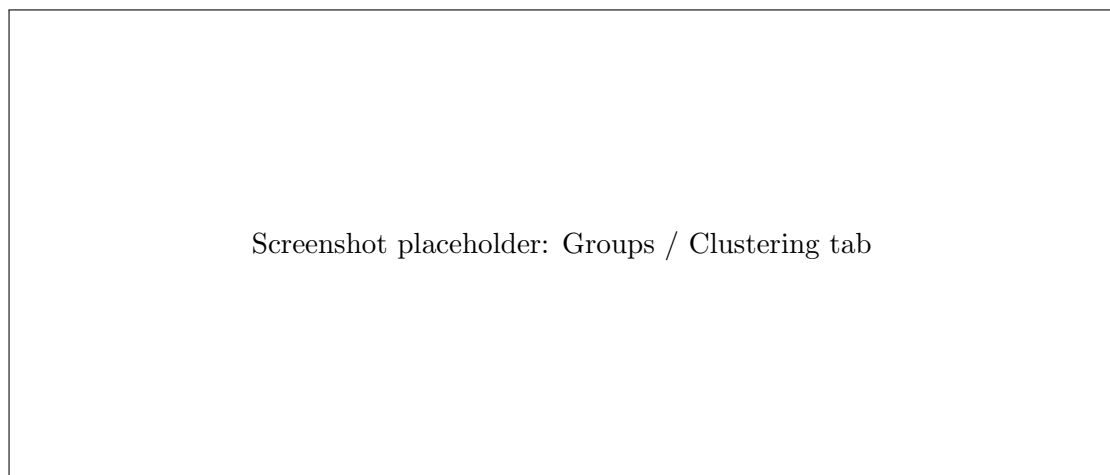


Figure 8.6: Groups / Clustering tab interface.

8.3.5 Volcano Plot

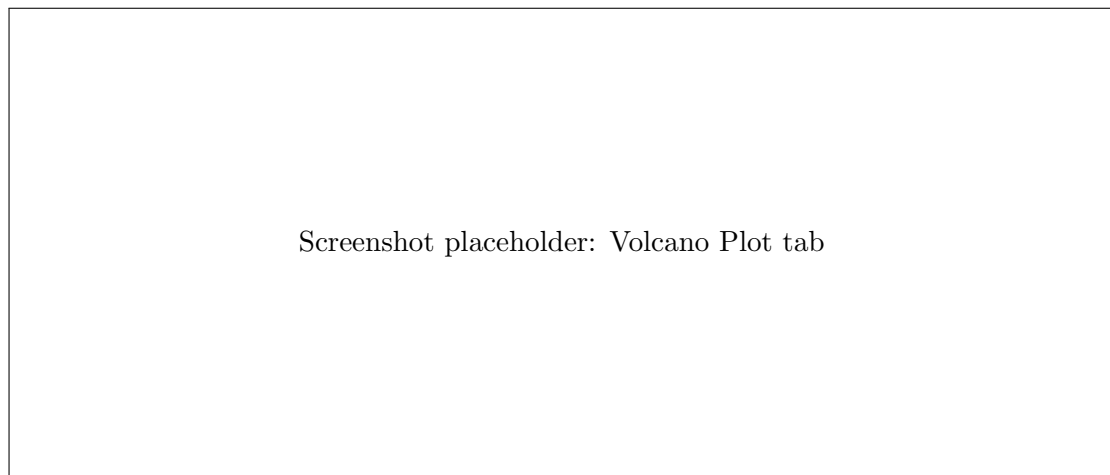


Figure 8.7: Volcano Plot tab interface.

8.4 Step 4 – Download your results

Most tests have a **Download CSV** button next to the results table.

8.5 Common mistakes

- Running a t-test or ANOVA without checking Normality first, on data that clearly is not normal – use the Wilcoxon/Kruskal-Wallis option instead in that case.
- Reading only the p-value and ignoring how large the actual difference is.
- Testing hundreds of variables on the Volcano Plot and trusting the raw p-value – use the FDR-adjusted option instead to avoid false positives.
- Assuming a strong correlation means one variable causes the other.

Chapter 9

Preprocessing Module

The Preprocessing module cleans up and transforms your data before modelling. It appears in the sidebar only after you click **Approve Dataset** in the Edit module, under three tabs: **Predictors**, **Outcome**, and **Data Preprocessed**.

9.1 Step 1 - Open the Predictors tab

This is where you preprocess your X (predictor) variables.

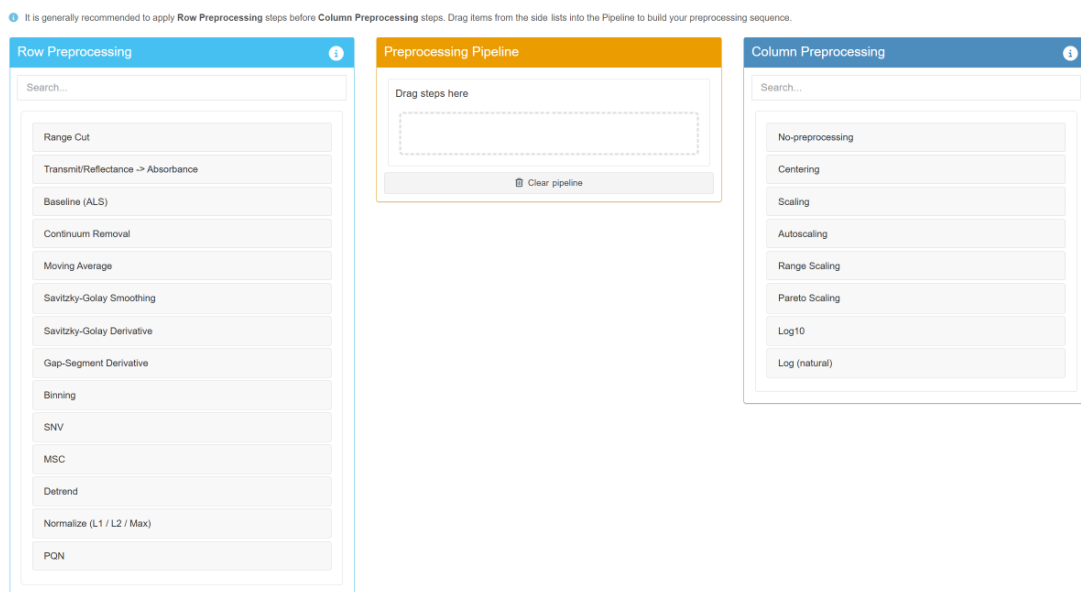


Figure 9.1: Predictors tab overview

9.2 Step 2 - Build your pipeline

Drag steps from the left and right lists into the middle **Pipeline** box. Reorder them by dragging, or click **Clear pipeline** to start over.

Tip

For spectral data is generally recommended to apply **Row Preprocessing** steps before **Column Preprocessing** steps. You can always come back and change it.

Step	What it does
Savitzky-Golay	Smooths noisy spectra (can also compute a derivative).
Moving Average	Simple smoothing.
Baseline (ALS)	Removes a slowly drifting background.
SNV / MSC	Corrects light-scattering effects between samples.
Detrend	Removes a curved baseline trend.
Normalize / PQN	Corrects for overall differences in signal strength between samples.
Range Cut	Keeps only a chosen range of columns, discards the rest.
Centering / Scaling / Autoscaling	Puts all variables on a comparable scale.
Log10 / Log (natural)	Compresses data that spans very different orders of magnitude.

A full description of every step, with formulas, is available from the info (book) icon on the tab.

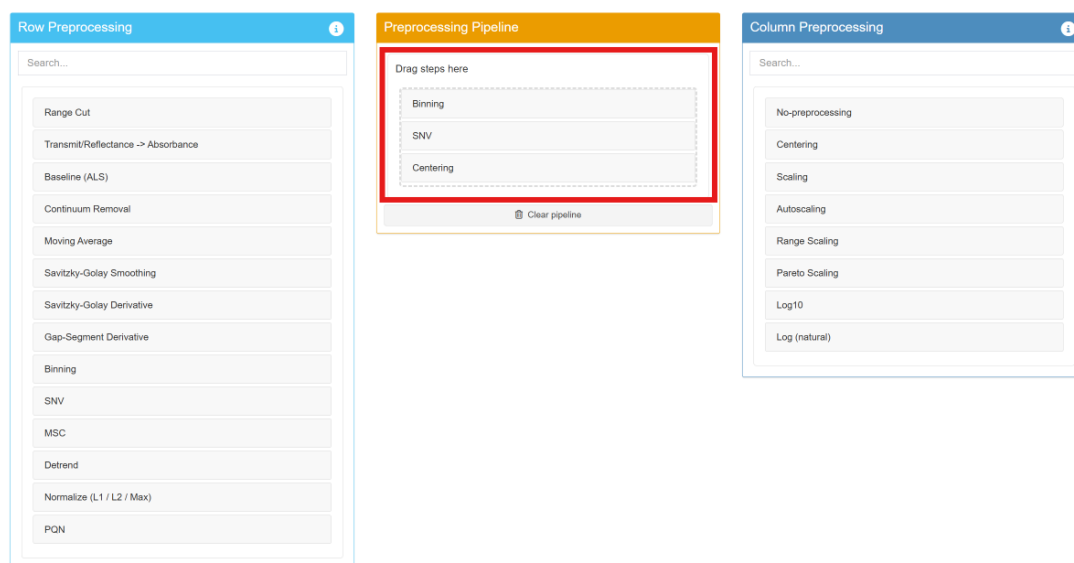


Figure 9.2: Dragging a step into the pipeline

9.3 Step 3 - Set each step's options (optional)

Once a step is in the pipeline, its settings panel appears below - most steps work fine with the default values.

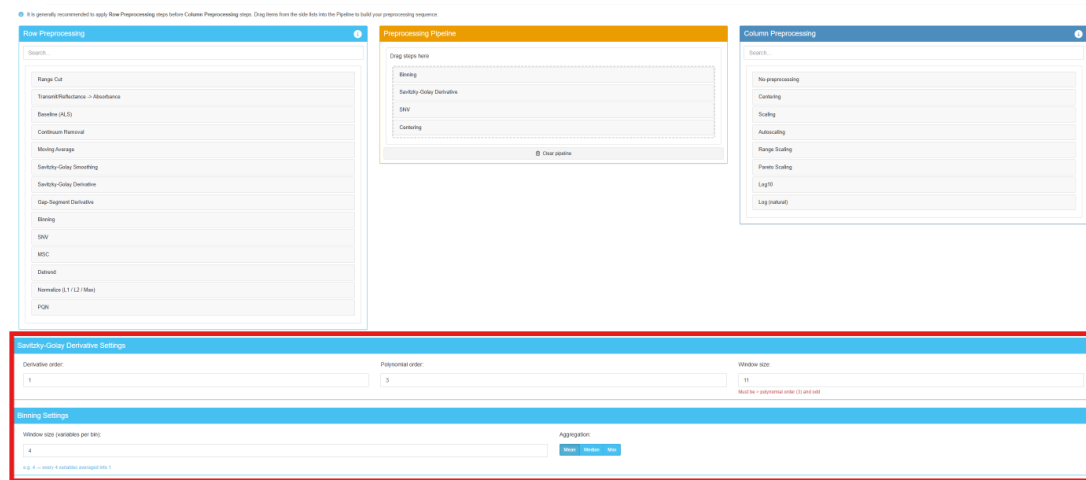


Figure 9.3: Step settings panel (e.g. Savitzky-Golay)

9.4 Step 4 - Preview the result

Click **Preprocess** to apply the selected preprocessing methods. The methods are applied sequentially, following the exact order in which they were added to the preprocessing pipeline. Each method therefore operates on the output produced by the previous one.

Once the preprocessing is complete, two charts are displayed side by side: **Original Data** and **Preprocessed Data**, allowing you to directly compare the dataset before and after the complete preprocessing sequence. The individual preprocessing steps are also listed below the charts in their order of application.

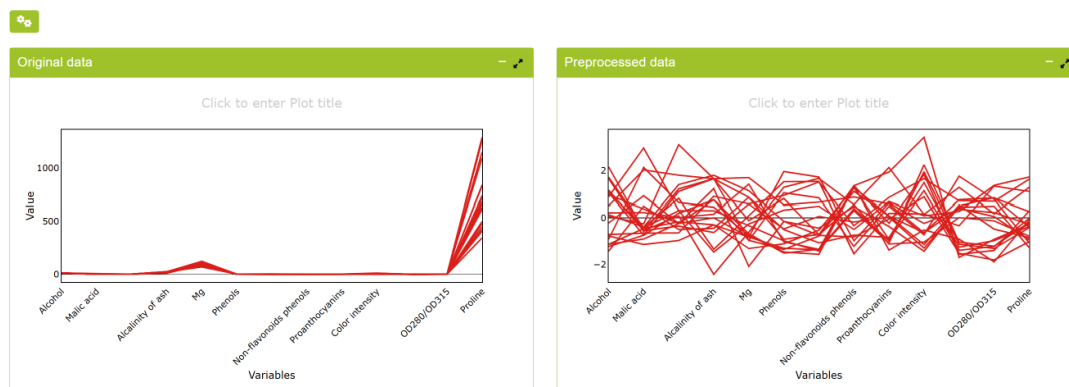


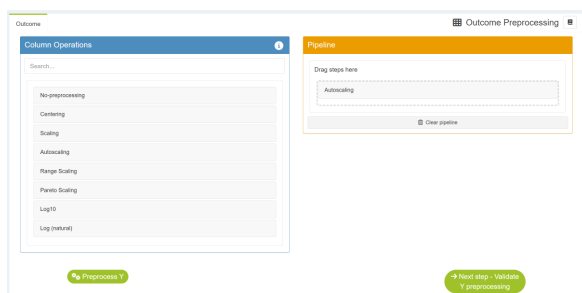
Figure 9.4: Original vs. Pre-processed comparison plots

9.5 Step 5 - Confirm and move on

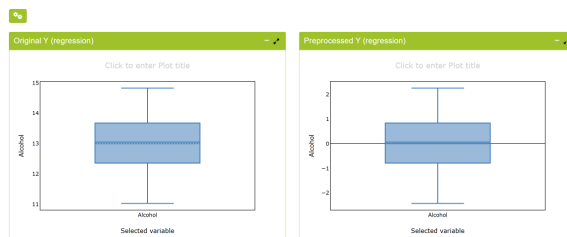
Happy with the result? Click **Next step - Validate preprocessing**.

9.6 Step 6 - Preprocess the Outcome (if you have one)

If you defined a regression target earlier, go to the **Outcome** tab and repeat the same drag-and-drop process, then click **Preprocess Y** and **Next step - Validate Y preprocessing**. If you did not define one, this tab is skipped automatically.



(a) Box preprocessing Y



(b) Outcome preprocessed after autocasting

9.7 Step 7 - Check the final result

The **Data Preprocessed** tab shows a read-only summary: the final data table and a count of samples/variables. Nothing can be changed here - go back to Predictors/Outcome to make edits.

Preprocessed Data											
Show 10 entries		Search:									
ID	Alcohol	Category	Data	Malic acid	Ash	Alcalinity of ash	Mg	Phenols	Flavanoids	Non-flavanoids phenols	
1 OLO1	1.520	Barolo	1995-01-07	-0.565	0.231	-1.166	1.909	0.807	1.032	-0.658	
2 OLO2	0.249	Barolo	1995-01-08	-0.502	-0.826	-2.484	0.018	0.567	0.732	-0.818	
3 OLO3	0.200	Barolo	1995-01-09	0.017	1.106	-0.268	0.088	0.807	1.212	-0.497	
4 OLO4	1.692	Barolo	1995-01-10	-0.350	0.487	-0.807	0.928	2.484	1.462	-0.979	
5 OLO5	0.299	Barolo	1995-01-11	0.224	1.835	0.451	1.278	0.807	0.661	0.226	
6 OLO6	1.483	Barolo	1995-01-12	-0.520	0.304	-1.286	0.858	1.558	1.362	-0.176	
7 OLO7	1.717	Barolo	1995-01-13	-0.422	0.304	-1.466	-0.262	0.327	0.491	-0.497	
8 OLO8	1.310	Barolo	1995-01-14	-0.171	0.888	-0.567	1.488	0.487	0.481	-0.417	
9 OLO9	2.259	Barolo	1995-01-15	-0.628	-0.716	-1.645	-0.192	0.807	0.952	-0.577	
10 OLO10	1.063	Barolo	1995-01-16	-0.888	-0.352	-1.047	-0.122	1.094	1.122	-1.140	
Showing 1 to 10 of 20 entries											
Previous 1 2 Next											
178 Samples			2 Factor/Classification variables			1 Regression variables			12 X variables		

Figure 9.6: Data preprocessed

9.8 Common mistakes

- Applying scaling steps before smoothing/correction steps - add Row steps first, then Column steps.
- Stacking several scaling steps together (e.g. Centering + Scaling + Autoscaling) - pick just one.
- Skipping the Outcome tab by accident when you actually have a regression target - double check it was detected in the Edit module.

Warning

Important. All preprocessing parameters are estimated exclusively from the **Training set**. The resulting values are then used to transform the **Test set**; no information from the Test set is used during the estimation step.

The preprocessing calculations performed in this section are used only for the **Visualization** and **Statistics** modules, where the preprocessed data must be generated in advance for plotting and statistical analysis.

For the **Unsupervised**, **Classification**, and **Regression** modules, the selected preprocessing sequence is not applied beforehand. Instead, it is fitted separately within each cross-validation iteration using only the corresponding training fold, and then applied to the validation fold. This procedure prevents data leakage and ensures an unbiased evaluation of model performance.

Warning

One-Class Modeling. In the **One-Class Modeling** module, preprocessing is handled directly within the module because the underlying procedure is conceptually different from standard classification and regression.

Preprocessing parameters are estimated exclusively from the samples belonging to the selected target class and are not calculated using the entire dataset. Further details about this procedure are provided in [Section 12](#).

Chapter 10

Unsupervised

The Unsupervised module explores the natural structure of your data without using any class or target column. It provides four techniques as separate sub-tabs: **PCA**, **HCA**, **K-means**, and **t-SNE**. The module becomes available once the preprocessing workflow has been defined.

A feature shared by all methods allows samples or variables to be removed directly after the exploratory evaluation, without restarting the entire analysis workflow. When a reduced dataset is created, the selected preprocessing steps are recalculated and applied to the updated data, rather than reusing the values previously computed on the original dataset. This ensures that the removed samples or variables do not influence the new preprocessing results and avoids introducing data leakage.

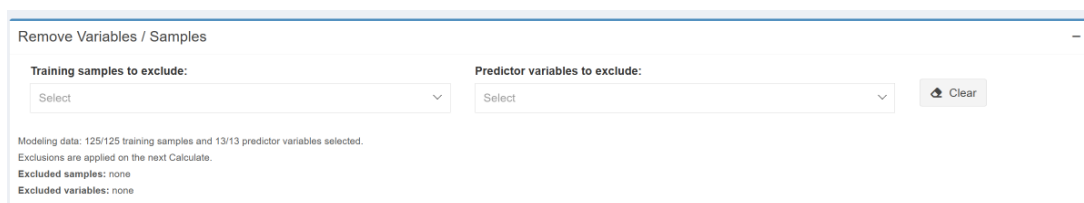


Figure 10.1: Remove samples or variables

10.1 PCA (Principal Component Analysis)

Use PCA to see your samples on a simple 2D/3D map and spot groups or outliers.

1. Set the **number of components**. The field is automatically pre-filled with **10 components**, or with a lower value when the dataset contains fewer than 10 predictor variables.
2. Select the PCA calculation algorithm from the dropdown menu: **SVD** or **NIPALS**.
3. Click **Calculate** to fit the PCA model using the selected settings.
4. Check the **Scree Plot** to inspect the variance explained by each component and the cumulative variance retained by the model.
5. Explore the **Scores Plot** and the corresponding **Loadings Plot**. The Scores Plot displays the samples in the selected component space, while the Loadings Plot shows the contribution of the predictor variables to those components. Use the available controls to change the components displayed in both plots.

- Open the **Hotelling's T^2 vs. Q Residuals** plot to inspect the samples according to their score and residual distances and to identify potentially unusual observations. When a sample lies outside the T^2 or Q acceptance limits, click its point to display the corresponding **Contribution Plot** and inspect which variables contribute most to its unusual behaviour.
- Click **Save PCA model** to store the fitted model and reuse it later in the **Predict** module.

Tip

As throughout the application, clicking the **gear-icon button** opens the plot settings. These options allow you to customise the graph, including its colours and point shapes, and to select the variables or samples to display.

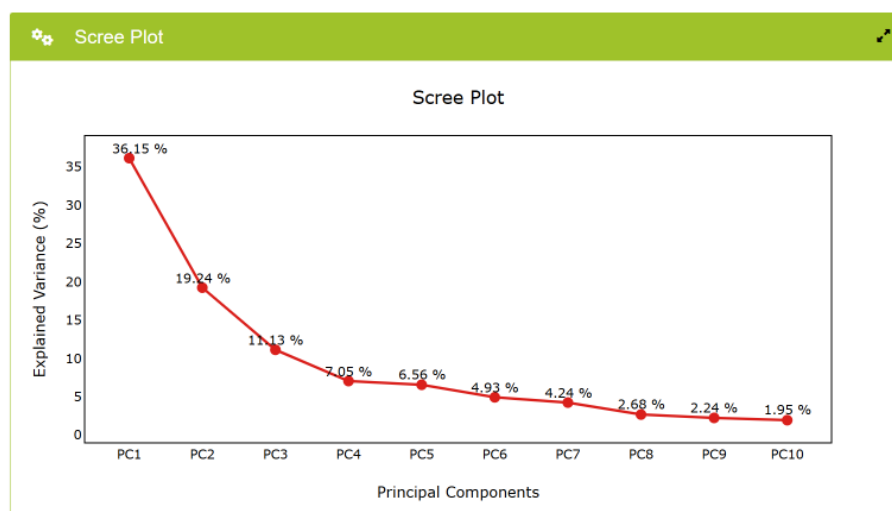


Figure 10.2: PCA Scree Plot

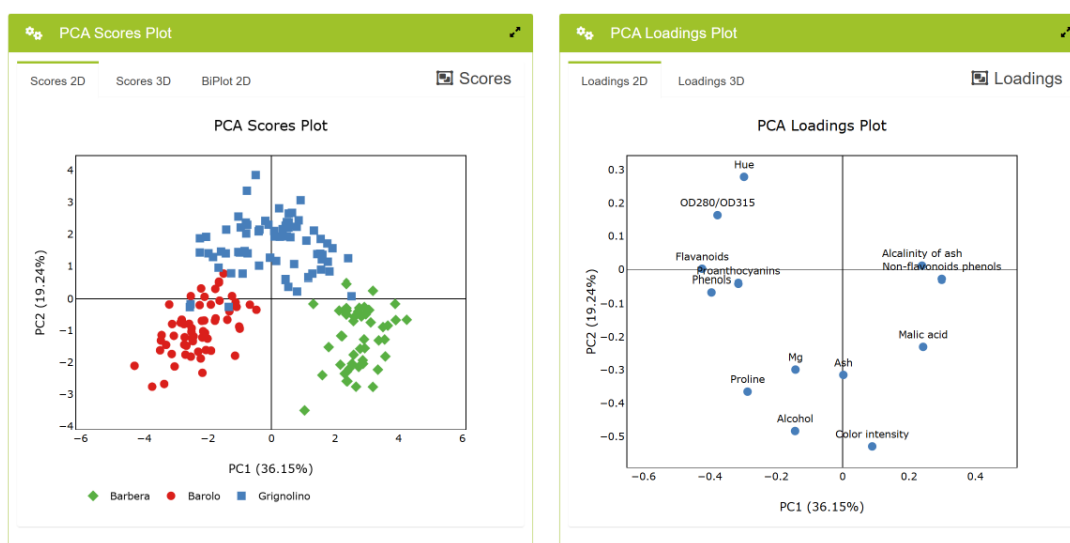


Figure 10.3: PCA Scores and Loading Plots coloured by a categorical variable



Figure 10.4: PCA Scores and Loading Plots coloured by a continuous variable and loadings plot in "Spectra mode"

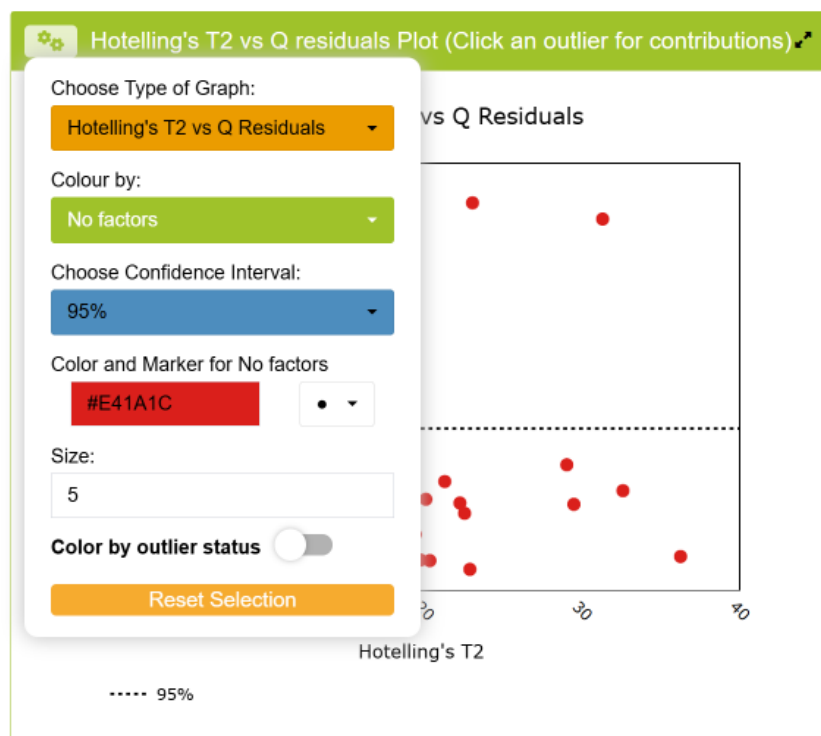


Figure 10.5: PCA Hotelling's T^2 vs. Q plot

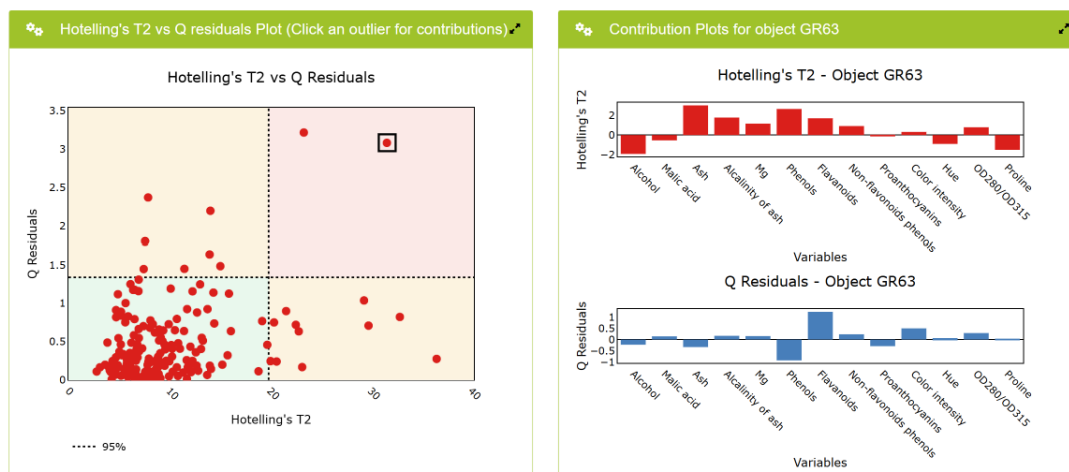


Figure 10.6: Contribution Plot

Tip

The **Contribution Plot** is available only for samples located outside the Hotelling's T^2 limit, the Q residual limit, or both limits. Samples that fall within both acceptance limits cannot be selected and no contribution results or plots are generated for them.

If you have a Test set, a **Test** tab appears once a model is trained: click **Run PCA projection** to see where the new samples fall.

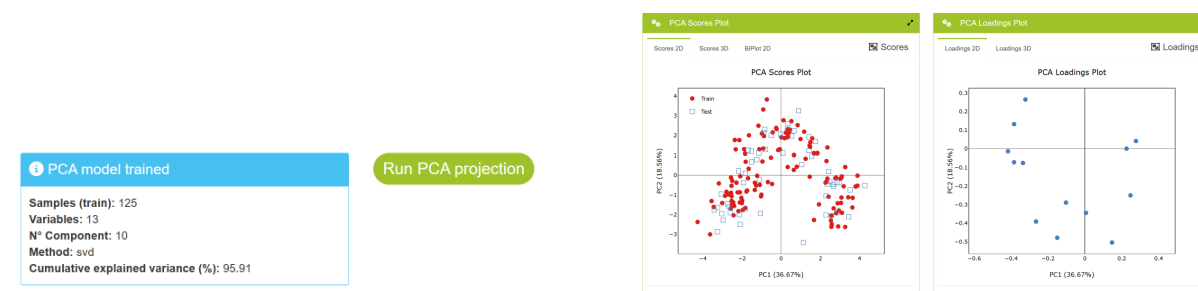


Figure 10.7: PCA test tab interface

10.2 HCA (Hierarchical Cluster Analysis)

Use HCA to group similar samples into clusters and display their relationships as a tree diagram.

1. Choose the **distance** and **linkage** methods.
2. Optionally select **Perform PCA first?** to run HCA on PCA scores instead of the original variables.
3. Set the **number of clusters**.
4. Click **Calculate**.
5. Inspect the **Dendrogram**: samples that merge at lower levels are more similar, while the selected clusters are highlighted in the final tree.

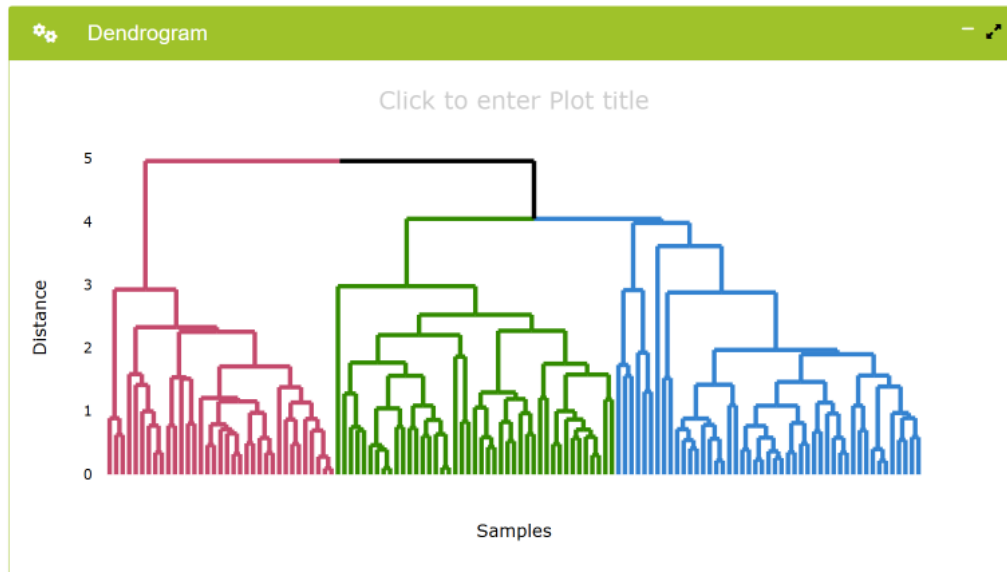


Figure 10.8: HCA Dendrogram

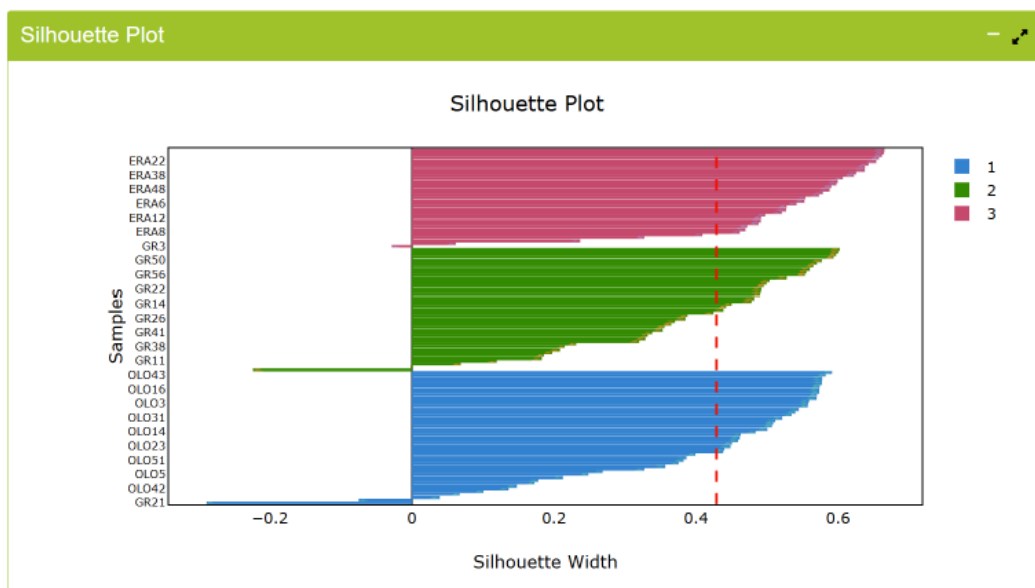


Figure 10.9: HCA Silhouette Plot



Figure 10.10: Comparison Cluster Analysis vs Original Data after PCA decomposition

10.3 K-means

Use K-means to group samples into a fixed number of clusters.

1. Optionally select **Perform PCA first?** to run K-means on PCA scores instead of the original variables.
2. Click **Calculate** to test several cluster numbers and obtain a suggested solution.
3. Adjust **Nr. of clusters** if a different grouping is preferred.
4. Click **Validate** to fit and display the final clustering result.

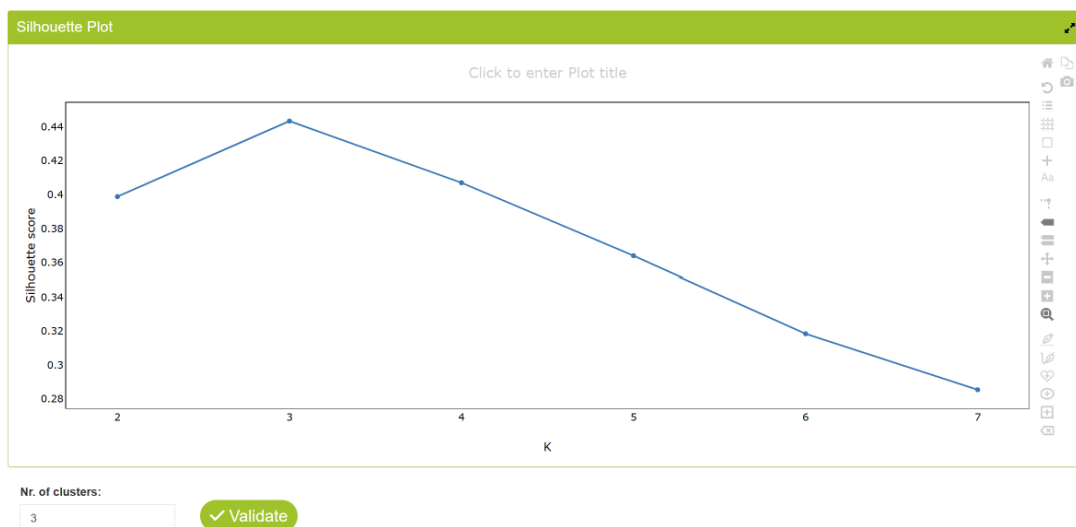


Figure 10.11: K-means Silhouette plot

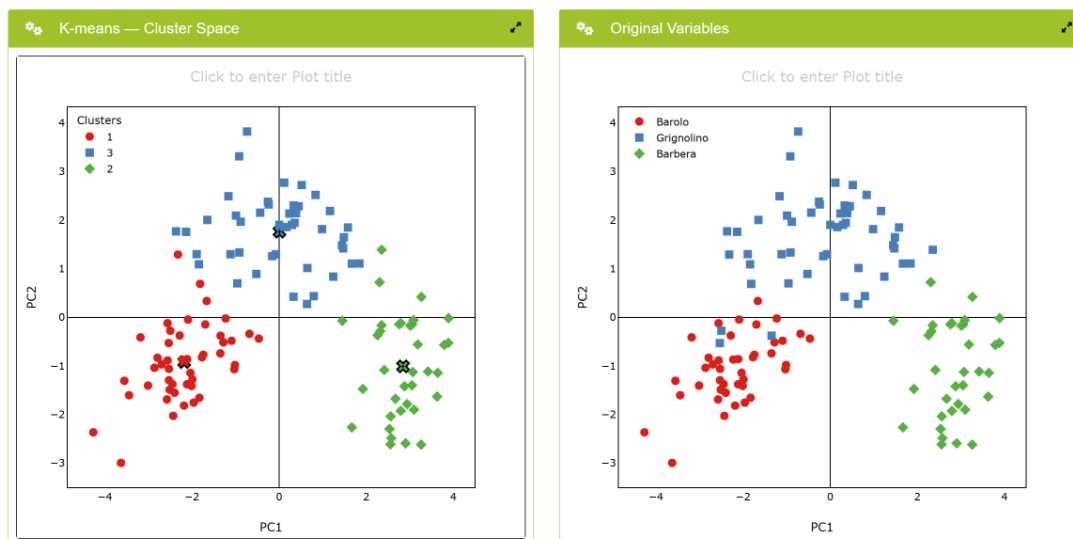


Figure 10.12: K-means Cluster Space plot

10.4 t-SNE (t-distributed stochastic neighbor embedding)

Use t-SNE for a more flexible 2D/3D map when PCA does not separate your samples well.

1. Open the **t-SNE settings** dropdown menu to configure the model parameters.
2. Set the number of **Dimensions** to 2 or 3 and adjust the remaining settings if needed.
3. Click **Calculate** to run t-SNE and display the resulting sample map.



Figure 10.13: t-SNE embedding plot

Warning

Only the distance between *neighbouring* points in a t-SNE plot is meaningful. Do not read too much into the distance between separate clusters.

Chapter 11

Classification Module

The Classification module builds a model that predicts a **category** (e.g. product type, authentic vs. counterfeit) from your data. It offers four methods: **KNN**, **LDA**, **PLS-DA** and **QDA**. Available once Preprocessing is complete.

Tip

Not sure which method to pick? **PLS-DA** is a good default for spectral/chemometric data with many predictor columns.

11.1 Step 1 – Pick a method and set Y

Open a method tab (KNN, LDA, PLS-DA or QDA) and select your classification (Y) variable.

Placeholder for Figure 11.1: Classification – method tab, Y selection interface.

Figure 11.1: Classification – method tab, Y selection interface.

11.2 Step 2 – (Optional) Set up cross-validation

Click **Cross-validation** to check how well the model will generalise to new data, instead of just fitting the training data. Pick a strategy (**Random k-fold** is a safe default), set the number of folds, then click **Confirm cross-validation** and **Apply**.

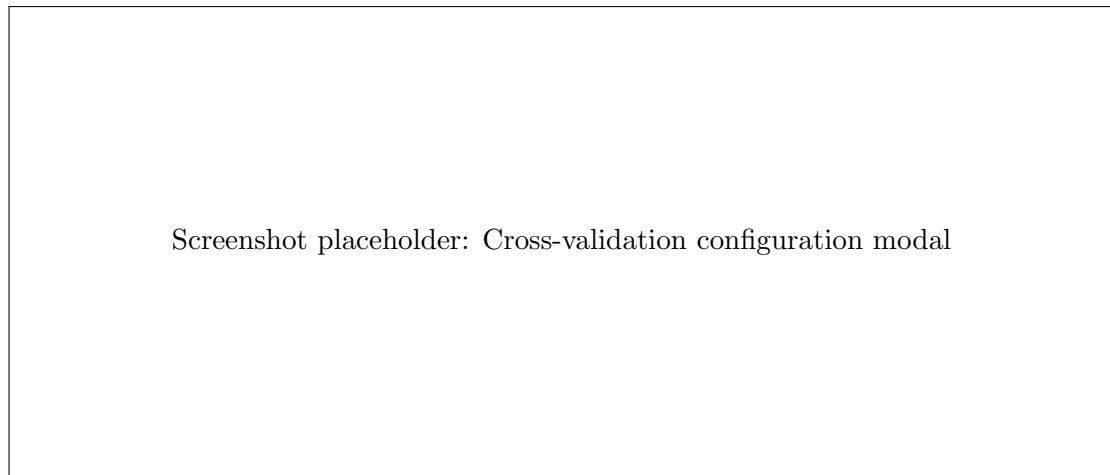


Figure 11.2: Cross-validation configuration modal interface.

11.3 Step 3 – Train the model

Click **Calculate**. For KNN, LDA and QDA, then click **Show Plots** (or **Plot Scores & Loadings**) to see the results.

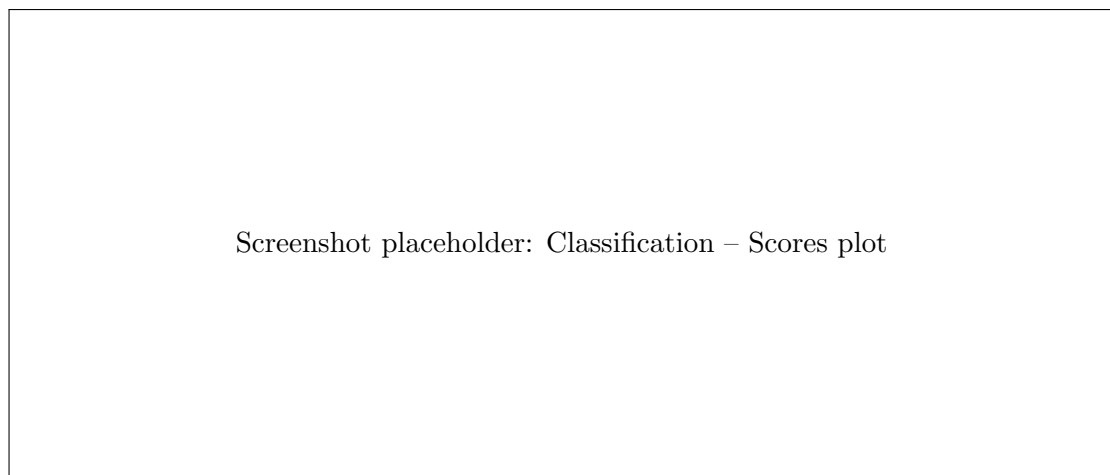


Figure 11.3: Classification – Scores plot interface.

11.4 Step 4 – Check the results

Look at the **Accuracy**, **Sensitivity** and **Specificity** numbers, and the confusion matrix (which classes get confused with which).

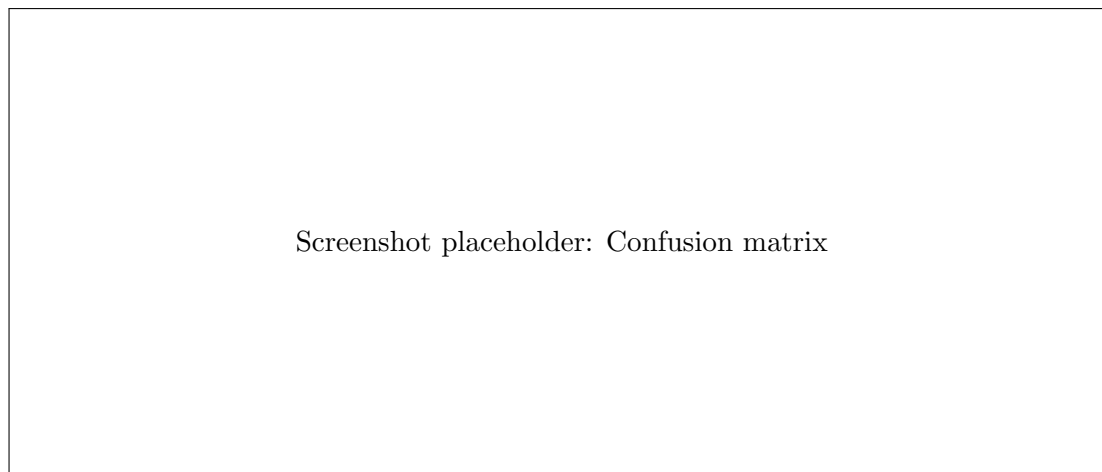


Figure 11.4: Confusion matrix interface.

11.5 Step 5 – Test on new data (if you have a Test set)

Once trained, a **Test** tab appears. Click **Run ... Projection** to apply the model to your Test rows and see the same results for unseen data.

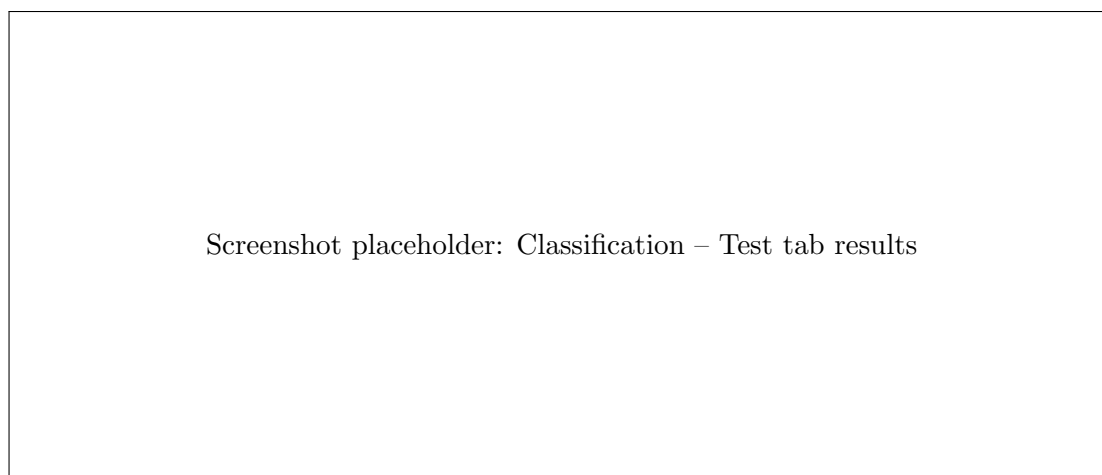


Figure 11.5: Classification – Test tab results interface.

11.6 Step 6 – Save the model

Use the download option on the Train tab to save the model so you can reuse it later in Predict (Chapter 14).

11.7 Common mistakes

- Running **Run ... Projection** before finishing Step 3 on the Train tab.
- Using **LDA** when you have more predictor columns than samples – turn on "Perform PCA first?" instead.
- Trusting a model's accuracy without checking its cross-validated score, not just the training score.

Note

A Logistic Regression engine exists in the underlying code but is not currently reachable from the interface, so it is not documented here.

Chapter 12

One Class Modeling Module (SIMCA)

One Class Modeling answers a different question than Classification: *"does this sample belong to class X?"* (accept/reject) rather than *"which class does it belong to?"*. It uses the SIMCA method.

Warning

SIMCA never compares classes to each other - a "Rejected" result only means the sample does not fit the target class, not that it belongs to another specific one.

12.1 Step 1 - Set Up the Target Class

Select the classification (**Y**) variable and the **target class** to be modelled.

Warning

SIMCA models only one class at a time. It does not automatically build one model for each available class; only a single model is created for the selected target class.

The screenshot shows the 'Setup' tab of the SIMCA module. At the top, there are two dropdown menus: 'Select Y variable:' with 'Category' selected, and 'Select target class:' with 'Barbera' selected. To the right of these is a green button labeled 'Accept & Continue →'. Below these is a section titled 'Predictors' with a sub-header 'Predictors Preprocessing'. A blue information icon is followed by the text: 'It is generally recommended to apply Row Preprocessing steps before Column Preprocessing steps. Drag items from the side lists into the Pipeline to build your preprocessing sequence.' The main area is divided into three panels. The 'Row Preprocessing' panel (blue header) contains a search bar and a list of steps: 'Range Cut', 'Transmit/Reflectance -> Absorbance', 'Baseline (ALS)', 'Continuum Removal', 'Moving Average', 'Savitzky-Golay Smoothing', and 'Savitzky-Golay Derivative'. The 'Preprocessing Pipeline' panel (orange header) contains a dashed box labeled 'Drag steps here' and a 'Clear pipeline' button. The 'Column Preprocessing' panel (blue header) contains a search bar and a list of steps: 'No-preprocessing', 'Centering', 'Scaling', 'Autoscaling', 'Range Scaling', 'Pareto Scaling', and 'Log10'.

Figure 12.1: One-Class Modeling - Setup tab

Note

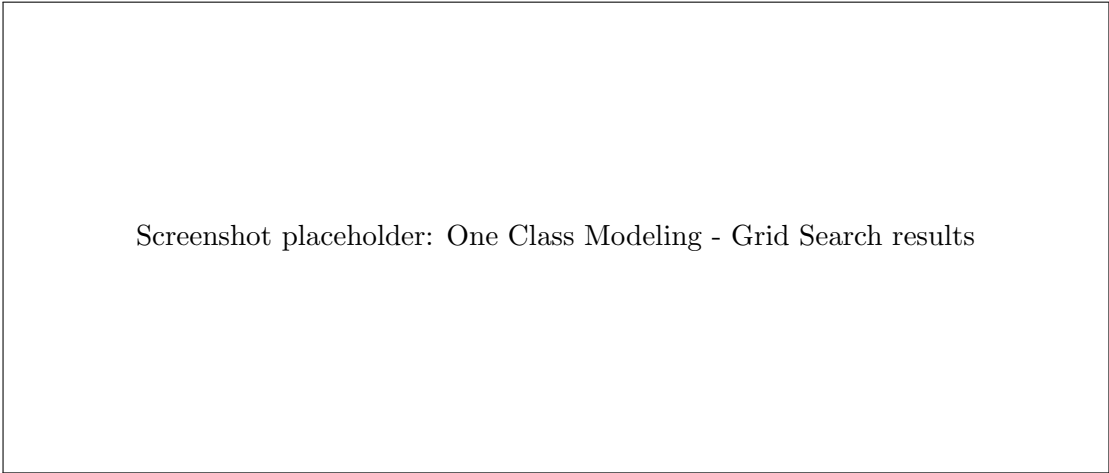
The preprocessing settings used in this module are independent of those configured in the main **Preprocessing** module. In one-class modelling, preprocessing must be estimated exclusively from the training samples belonging to the target class. Configure the required preprocessing steps again in this tab, then click **Accept & Continue**.

12.2 Step 2 - (Optional) Set up cross-validation

Click **Cross-validation** to check how well the model generalises. See Section 11.2 for the available strategies.

12.3 Step 3 - Train the model

Set **Max components** and **Alpha** (a smaller alpha is stricter), then click **Calculate**. If you set up cross-validation, review the suggested number of components and click **Fit Best Model**.

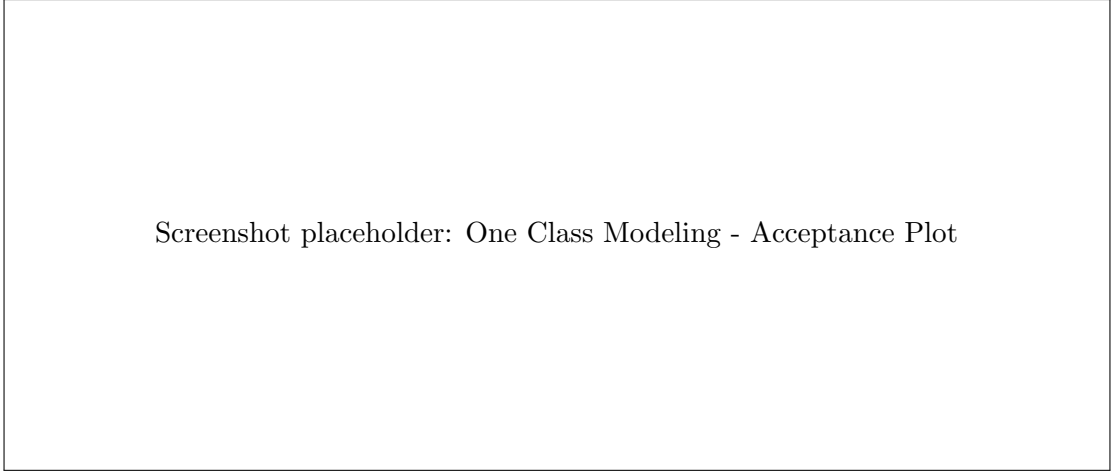


Screenshot placeholder: One Class Modeling - Grid Search results

Figure 12.2: One Class Modeling - Grid Search results interface.

12.4 Step 4 - Check the results

The **Acceptance Plot** shows which samples fall inside (accepted) or outside (rejected) the target class boundary. Check **Efficiency**, **Sensitivity** and **Specificity** below it.

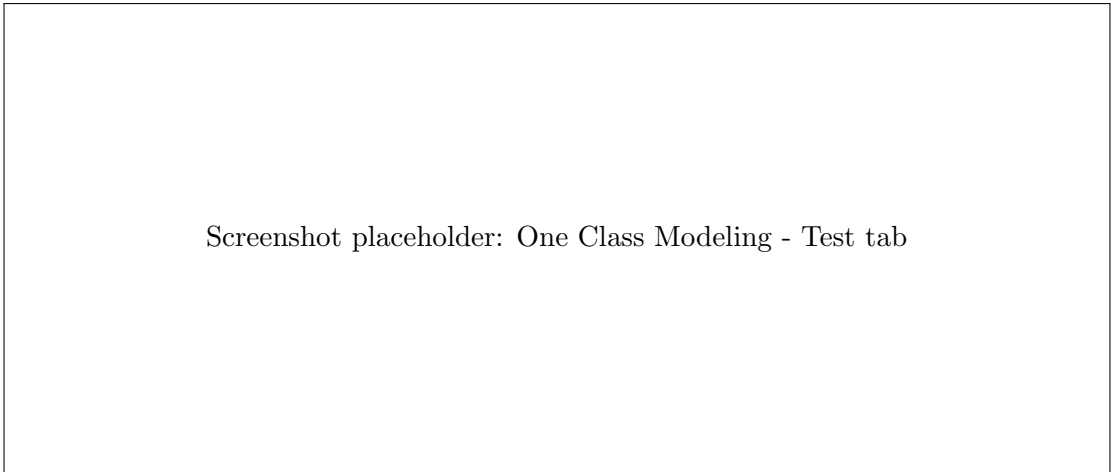


Screenshot placeholder: One Class Modeling - Acceptance Plot

Figure 12.3: One Class Modeling - Acceptance Plot interface.

12.5 Step 5 - Test on new data (if you have a Test set)

Once trained, a **Test** tab appears. Click **Run SIMCA Projection** to apply the model to your Test rows.



Screenshot placeholder: One Class Modeling - Test tab

Figure 12.4: One Class Modeling - Test tab interface.

12.6 Step 6 - Save the model

Click **Save SIMCA model** to reuse it later in Predict (Chapter 14).

12.7 Common mistakes

- Treating a "Rejected" result as if it identified a different class - it only means "outside the target class".
- Choosing too few or too many components relative to your number of target-class samples.
- Forgetting that global Preprocessing settings do not apply here.

Chapter 13

Regression Module

The Regression module builds a model that predicts a **number** (e.g. a concentration) from your data. It offers three methods: **MLR**, **PCR** and **PLS**. Available once Preprocessing is complete.

💡 Tip

Not sure which method to pick? **PLS** is a good default for spectral/chemometric data with many predictor columns.

13.1 Step 1 – Pick a method and set Y

Open a method tab (MLR, PCR or PLS) and select your numeric (Y) variable to predict.

Screenshot placeholder: Regression – method tab, Y selection

Figure 13.1: Regression – method tab, Y selection interface.

13.2 Step 2 – Set the number of components (PCR/PLS only)

For PCR and PLS, set the **max number of components/latent variables** to try – Cactus will pick the best one for you using cross-validation. MLR has no such setting.

13.3 Step 3 – (Optional) Set up cross-validation

Click **Cross-validation** to check how well the model generalises to new data. See Section 11.2 for the available strategies. A **Random k-fold** is a safe default.

13.4 Step 4 – Train the model

Click **Calculate**, then (for PCR/PLS) **Plot Scores, Loadings & Scree** to see the results.

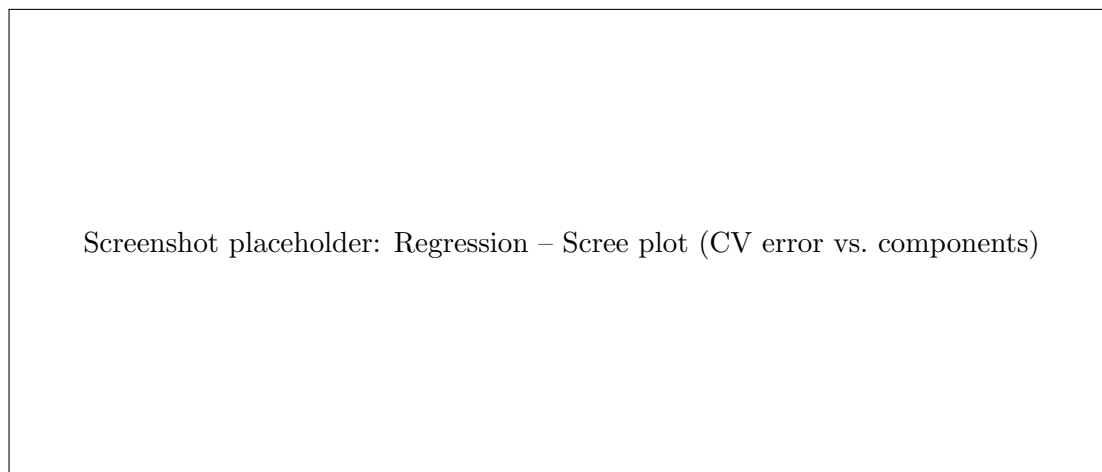


Figure 13.2: Regression – Scree plot (CV error vs. components) interface.

13.5 Step 5 – Check the results

Look at **RMSE** and **R²**: lower RMSE and higher R² mean a better fit. For MLR, also check the **VIF** tab for warning signs of redundant predictors.

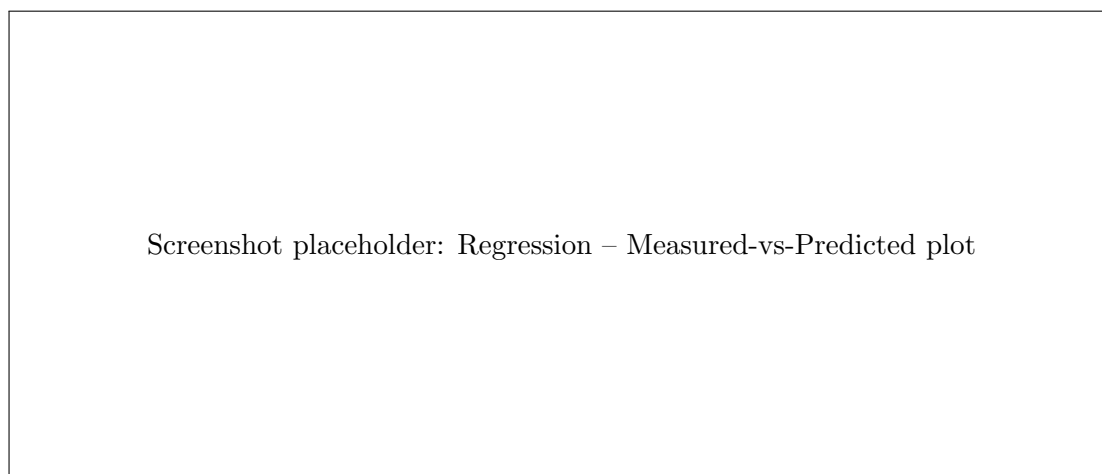
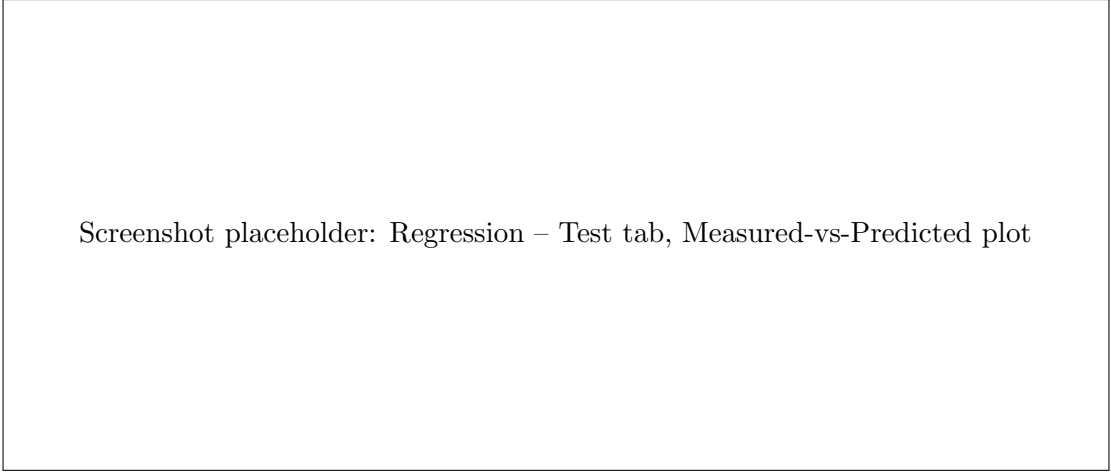


Figure 13.3: Regression – Measured-vs-Predicted plot interface.

13.6 Step 6 – Test on new data (if you have a Test set)

Once trained, a **Test** tab appears. Click **Run Predictions ...** to apply the model to your Test rows.



Screenshot placeholder: Regression – Test tab, Measured-vs-Predicted plot

Figure 13.4: Regression – Test tab, Measured-vs-Predicted plot interface.

13.7 Step 7 – Save the model

Use the download option on the Train tab to save the model so you can reuse it later in Predict (Chapter 14).

13.8 Common mistakes

- Choosing too many components for PCR/PLS – watch the Scree plot; more is not always better past a certain point.
- Using MLR with many correlated predictors and few samples – check the VIF tab, or use PCR/PLS instead.
- Trusting a model's fit on training data without checking its cross-validated performance.

Chapter 14

Predict Module

The Predict module lets you apply a model you already saved to new data, without repeating the whole Import/Edit/Preprocess/Train pipeline. It has two sidebar sub-items: **Upload model** and **Prediction**.

14.1 Step 1 – Upload your saved model

Click **Upload saved model (.rds)** and select the model file you downloaded earlier from a Train tab. A summary appears showing the model type and its saved metrics.

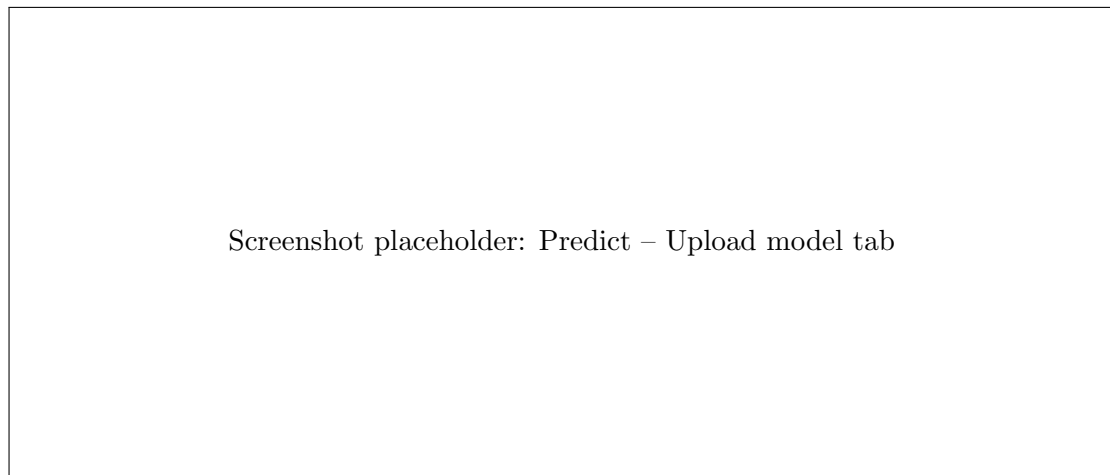


Figure 14.1: Predict – Upload model tab interface.

14.2 Step 2 – Upload your new data

In the **Load Test Data** box, choose your file type and upload the new data (same options as the Import module), then click **Import**.

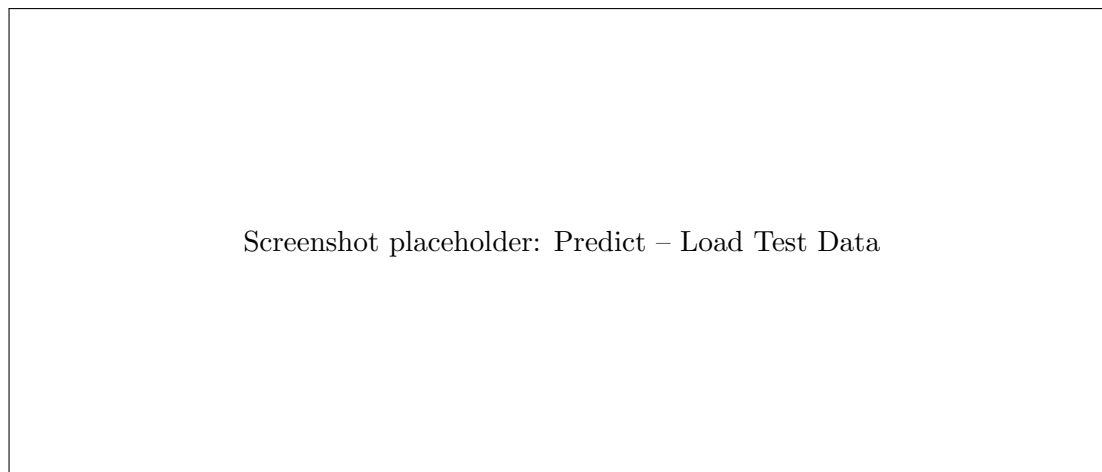


Figure 14.2: Predict – Load Test Data interface.

14.3 Step 3 – Check the Schema tab

Cactus compares your new data's columns against what the model expects. Fix anything flagged in red (missing required columns) before continuing; for missing values, pick an imputation method on the **Missing Values** tab.

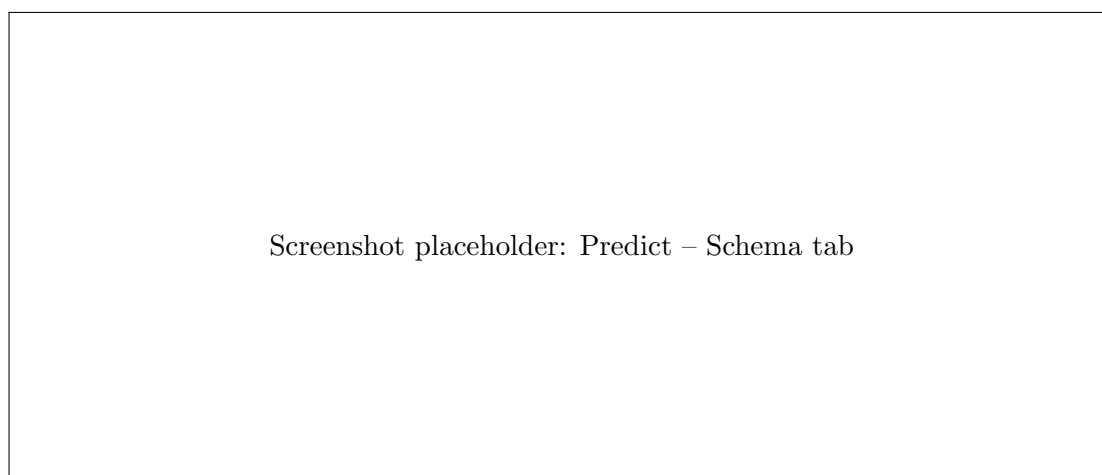


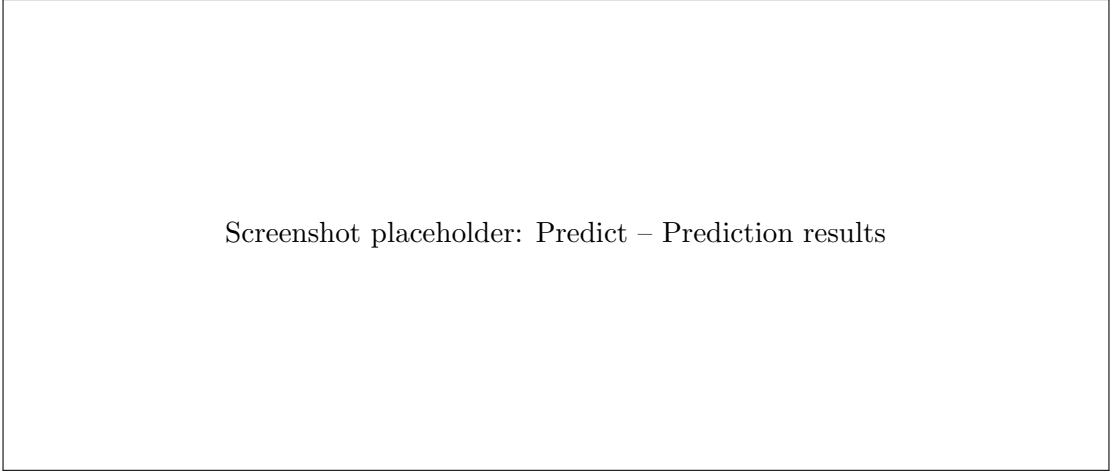
Figure 14.3: Predict – Schema tab interface.

14.4 Step 4 – Run the prediction

Click **Run Prediction**.

14.5 Step 5 – Review and download results

Results depend on the model type: classification models show predicted classes, regression models show predicted values, PCA/SIMCA show where the new samples fall relative to the trained model. Every results table has **CSV/Excel** download buttons.



Screenshot placeholder: Predict – Prediction results

Figure 14.4: Predict – Prediction results interface.

14.6 Common mistakes

- Uploading data whose columns do not match the model’s predictors – Cactus lists exactly which columns are missing.
- Leaving missing values in required columns without picking an imputation method.
- Uploading the wrong model file (e.g. a regression model’s predictors for a classification task) – this shows up as a missing-column error, so double-check which file you uploaded.
- Uploading a file that is not `.rds`, or a corrupted one.

Chapter 15

Credits

15.1 Purpose

The Credits tab (sidebar entry **Credits**) displays information about the people and organisations behind Cactus. It has no inputs, parameters, or interactive analysis – it is a static information page.

15.2 What is shown

- The application logo and a short description naming the development team, and the line "Built with R / Shiny, with the support of DataBloom" (with R, Shiny and DataBloom logos).
- A feedback box linking to a Google Form for sending feedback or reporting a bug.
- Team member cards, each with a photo, name, role and affiliation:

Name	Role / Affiliation
Eugenio Alladio, PhD	Associate Professor, Department of Chemistry, University of Turin
Alberto Mazzoleni	PhD Student, Department of Chemistry, University of Turin
Giovanni Solarino	PhD Student, Department of Chemistry, University of Turin
Lorenzo Castellino	PhD Student, Department of Chemistry, University of Turin
Alessandra Olarini	Research Fellow, Department of Chemistry, University of Turin
Marco Pazzi	Associate Professor, Department of Chemistry, University of Turin
Nicola Cavallini	Assistant Professor, DISAT, Politecnico di Torino

At the bottom of the page, two institutional logos link out: "Dipartimento di Eccellenza" and DataBloom.

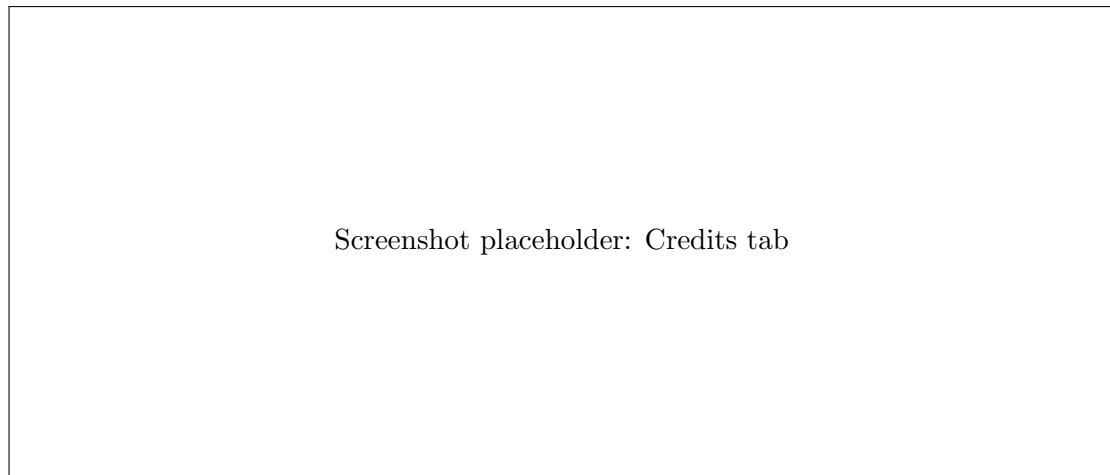


Figure 15.1: Credits tab interface.

i Note

This page is informational only – there is nothing to configure or troubleshoot here. If you need to report a problem with the application, use the feedback link on this tab.

Chapter 16

Plots and Visual Outputs

Every chart in Cactus - from a simple boxplot to a PCA scores plot - is built with the same interactive plotting engine (Plotly) and shares one common toolbar and set of gestures. This chapter documents that shared behaviour once, and then indexes every plot type produced by the application and where to find its detailed interpretation guidance.

16.1 Interacting with any plot

16.1.1 Toolbar buttons (right-hand side of every plot)

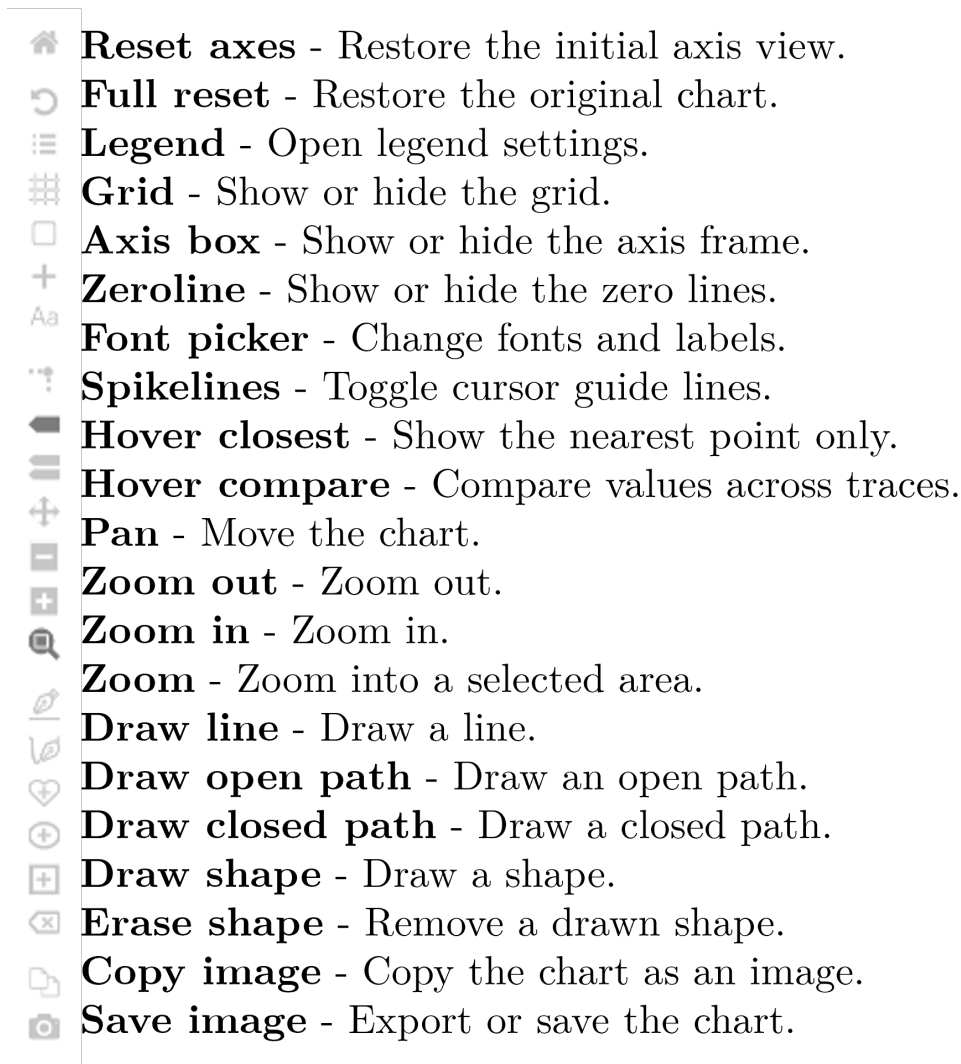


Figure 16.1: Toolbar and interactivity of Plotly

Tip

Titles, axis titles, legend text and annotation text can be edited directly by clicking on them in the plot (Plotly's "editable" text mode).

Note

For very large plots (tens of thousands of points), some hover/spike-line buttons are automatically dropped and tooltips may be disabled to keep the plot responsive. Dark mode (Section 3.3) automatically restyles every plot's background, font and axis colours.

16.2 Plot catalog

The table below lists every plot type produced by Cactus, grouped by module, with a pointer to the chapter that explains its required data, options and interpretation in full.

Module	Plot types	Full details
Visualization - One Variable	Boxplot, Histogram, Violin Plot, Density Chart, Bar Chart, Pie Chart	Chapter 7
Visualization - Two Variables	Scatter Plot, Bubble Plot, 2D Histogram, 2D Density Chart, Scatter Matrix, Time Series	Chapter 7
Visualization - Multiple Variables	3D Scatter Plot, Boxplots (multi), Heatmap, Spectra / Parallel Coordinates	Chapter 7
Statistics	Q-Q plot (Normality), Box plot with significance brackets (t-test, ANOVA), Correlation Heatmap, Clustered Heatmap, Volcano Plot	Chapter 8
Preprocessing	Original vs. Pre-processed comparison plots (Predictors and Outcome tabs)	Chapter 9
Unsupervised PCA	Scree Plot, Scores Plot (2D/3D/BiPlot), Loadings Plot, Hotelling's T^2 vs. Q Residuals Plot, Contribution Plots	Chapter 10
Unsupervised HCA	Dendrogram, Silhouette Plot, CA/Original data plots	Chapter 10
Unsupervised - K-means	Silhouette-score-vs-K curve, Cluster Space plot, Original Variables plot	Chapter 10
Unsupervised - t-SNE	t-SNE Scores Plot, Feature Coloring plot	Chapter 10
One Class Modeling (SIMCA)	Acceptance Plot (Score Distance vs. Orthogonal Distance), Contribution Plots, Scores/Loadings/Scree plots, confusion-matrix heatmap	Chapter 12
Classification KNN	Decision-region scatter plot	Chapter 11
Classification LDA / QDA	Scores plot, Loadings plot, Class Probability plot, ROC curves	Chapter 11
Classification PLS-DA	Scores/Loadings/Weights plots, VIP plot, Scree plot (CV accuracy)	Chapter 11
Regression - MLR	Residuals plot, Influence (Cook's-distance) plot	Chapter 13
Regression - PCR	PC Scores plot, Loadings plot, Scree plot (CV metric)	Chapter 13
Regression - PLS	Scores/Loadings/Weights plots, VIP plot, Leverage plot, Scree plot, Measured-vs-Predicted plot, residuals histogram	Chapter 13

Module	Plot types	Full details
Predict	Same plot families as the originating model type (PCA, SIMCA, KNN, LDA/QDA, PLS-DA, MLR/PCR/PLS), recomputed for the new data	Chapter 14

16.3 General interpretation guidance

- **Scores plots** (PCA, LDA/QDA, PLS-DA, PCR, PLS) place each sample in a reduced-dimension space; nearby points behave similarly under the model, and the axis labels/percentages tell you how much information that view actually captures.
- **Loadings plots** show how much each original variable contributes to a given component/latent variable - read them together with the corresponding scores plot, not in isolation.
- **Distance/limit plots** (PCA's T^2/Q , SIMCA's Acceptance Plot) always compare a statistic against a threshold line - points beyond the line are flagged as unusual or rejected; clicking a flagged point often reveals a contribution plot explaining *which* variables are responsible.
- **Measured-vs-Predicted plots** (Regression) should be judged against the 1:1 (perfect prediction) line, not only against the fitted trend line - a fit line close to 1:1 with points tightly clustered around it indicates a good model.
- **Confusion matrices** (Classification, SIMCA) show actual vs. predicted class counts; a strong diagonal indicates good performance, and off-diagonal cells identify which classes are being confused with which.

Warning

No single plot proves a model is good. Always read a plot together with its accompanying metrics table (accuracy/RMSE/ R^2 /etc.) and, where available, its cross-validated (not just calibration) values - calibration-only performance is typically optimistic.

Chapter 17

Exporting Results

Cactus does not have one central "export" screen; instead, every table and every trained model offers its own download control, right where it is produced. This chapter consolidates all of them in one place for quick reference.

17.1 Data tables

- **Import Data** preview – **Download CSV** (settings dropdown above the table).
- **Split Train/Test** – **Download TRAIN** / **Download TEST** (CSV), once a split has been run.
- **Edited Data** preview – **Download CSV** (settings dropdown).
- **Data Preprocessed** – **Download CSV** of the final combined (factor + X + Y) preprocessed dataset.
- **Statistics** tabs (t-test, ANOVA, Correlation Heatmap, Groups/Clustering, Volcano Plot) – each results table has its own **Download CSV** button.
- **Cross-validation** preview (Classification, One Class Modeling, Regression) – the fold row-assignment preview can be downloaded as Excel before you apply it.
- **Predict** – every predictions table (all model types) has **Excel** and **CSV** export buttons; the imported new-data preview also has its own **Download CSV** button.

17.2 Trained models

- **PCA** (Train and Test) – **Save PCA model** (.rds, reusable in the Predict module) and **Save PCA data** (.xlsx: variance, scores, loadings, Q/T²).
- **HCA** – **Save HCA model** (.rds).
- **t-SNE** – **Save t-SNE** (embedding table and parameters used).
- **One Class Modeling (SIMCA)** – **Save SIMCA model** (.rds).
- **Classification** (KNN, LDA, PLS-DA, QDA) and **Regression** (MLR, PCR, PLS) – each trained model can be saved as an .rds bundle for later reuse in the Predict module (Chapter 14).

 Tip

A saved `.rds` model bundle is the only file format the Predict module accepts. Keep these files together with a note of which dataset and preprocessing pipeline produced them, since the Predict module checks new data against the exact predictor columns the model was trained on.

17.3 Plots

Every plot can be exported as an image via its own toolbar (**Copy Image** / **Save Image**, or Ctrl/Cmd+C) – see Chapter 16 for the full list of plot interactions.

 TODO

No PDF or combined "full report" export was found anywhere in the application. If you need a single document combining multiple tables/plots, export each one individually (CSV/Excel/image) and assemble them outside Cactus.

Chapter 18

Troubleshooting

This chapter lists problems users are likely to encounter, grouped by symptom, with practical fixes. If your problem is not listed here, check the chapter for the specific module you were using – each one lists its own common mistakes.

18.1 Application will not launch

- **Console window closes immediately** – run `start_app.bat` from a command prompt (instead of double-clicking) so the error message stays visible, and check that the `R-4.5.2\` folder is present alongside `app.R`.
- **Browser does not open automatically** – open it manually and navigate to `http://127.0.0.1:3838` (or the port shown in the console).
- **Port already in use** – another instance of Cactus (or another application) is already using that port; close it, or launch with a different port: `start_app.bat 4000`.

18.2 Missing packages / R errors on startup

- The console prints `[CACTUS ...] ...loaded` messages as packages load; if it stops part-way through with an R error instead of reaching `"helpers + custom steps sourced -- global.R COMPLETE"`, a required package is missing or broken in the bundled R installation.

⚙️ TODO

This guide cannot provide exact package-installation commands, since the install scripts referenced in `start_app.bat`'s comments were not found in the current copy of the repository. Contact your administrator or the development team (Chapter 15) with the exact error message shown in the console.

18.3 Import problems

- **"File type mismatch"** – the selected **Type of file** does not match the uploaded file's extension; pick the matching type.
- **Wrong file format / garbled data in the preview** – check the **Delimiter** and **Decimal symbol** advanced options; files exported with European conventions (semicolon

delimiter, comma decimal) need those settings adjusted.

- **Duplicate values in first column** – disable **Use first column as row labels**, or make the identifiers in that column unique before importing.
- **Column mismatch** (Split module’s external test-set upload, or Predict module’s new data) – the uploaded file is missing a column that the reference dataset/model expects; add the missing column or use a file that matches the original schema.

18.4 Non-numeric variables where numeric ones are expected

- A column intended to be numeric appears as text if the source file mixes numbers with units, symbols, or inconsistent decimal separators – clean the column in the source file and re-import, or adjust the **Decimal symbol** option.
- Modelling tabs (Classification, Regression, Unsupervised) only use numeric columns as predictors; if a variable you expect to see as a predictor is missing, confirm in the Edit Pipeline (Chapter 6) that it was actually assigned the **X (predictor)** role and is numeric.

18.5 Missing values

- If the **Confirm imputation & proceed** button in the Missing Values tab stays disabled, some missing values are still unresolved – check the positions table and continue imputing.
- If a column has a very high percentage of missing values (shown in red in the overview table), consider removing it entirely in the Edit Pipeline rather than imputing it.
- **KNN imputation** cannot be applied to classification, regression or date columns – these are excluded automatically.
- **Kennard-Stone / DUPLEX split** cannot run while predictor data still has missing values – impute first, or choose a different split method.

18.6 Too many variables / too few samples

- **LDA** is automatically disabled with a warning when the number of numeric predictors is greater than or equal to the number of samples – enable the **Perform PCA first?** option to reduce dimensionality before running LDA.
- **QDA** needs enough samples per class to estimate a separate covariance matrix per class; very small classes, or high dimensionality without a PCA pre-step, can fail or produce unstable fits.
- **MLR** with many correlated predictors and few samples is prone to multicollinearity (check the **VIF** tab) – consider PCR or PLS, which handle this better via component reduction.
- **One Class Modeling (SIMCA)** with too few target-class training samples relative to the number of components chosen will overfit – reduce the number of components or gather more target-class samples.

18.7 Plot not appearing / looks empty

- Some plots only render after an explicit action, not automatically: click **Calculate/Preprocess/Plot** (e.g. the Scatter Matrix in Visualization does not auto-update on setting changes – click **Plot** again).
- A sidebar tab showing "complete the previous step" instead of a plot means an earlier pipeline stage (Approve Dataset, Preprocessing) has not been completed yet – see Chapter 3.
- A chart type that used to be visible but has disappeared from a menu may have been hidden via **Configure App** in the right-hand settings panel (Section 3.3) – check there before assuming something is broken.
- Very large plots may drop some interactive features (hover tooltips, certain toolbar buttons) automatically to stay responsive – this is expected behaviour, not an error.

18.8 Model will not run

- **"Complete the Preprocessing step first"** – Unsupervised, Classification, One Class Modeling and Regression all require Preprocessing (Chapter 9) to be finished first.
- **Run ...Projection / Test tab missing** – the Test tab for any modelling method only appears once a model has been trained *and* Test-split rows exist; go back to Split Train/Test (Chapter 6) if you need a Test set.
- **"Run KNN Projection" blocked** – click **Show Plots** on the Train tab first.
- **Cross-validation error about a missing class in a fold** – Venetian Blinds/Contiguous Blocks folds require ordered/sufficiently represented class data; try a Stratified or Random k-fold strategy instead, or reduce the number of folds.
- **Predict module: "required predictor column(s) missing"** – your new data does not contain every column the model was trained on; see Chapter 14.

18.9 Slow / unresponsive interface

- Checking **Show entire dataset** on a very large table will prompt a confirmation warning first, since it can slow down the browser – prefer paging through a limited number of rows/columns instead.
- Heatmaps and Spectra/Parallel Coordinates plots on very large datasets are rendered as a single rasterised image for performance; reduce the number of rows/columns/axis labels shown if the browser feels sluggish.

Chapter 19

Appendix

19.1 Glossary

Term	Practical meaning
Predictor (X) variable	An input measurement used to explain or predict something else (e.g. a spectral intensity at one wavelength).
Classification (Y) variable	A categorical column used as a class label to discriminate between groups.
Regression (Y) variable	A numeric column used as the target a regression model tries to predict.
Train / Test split	Dividing samples into a set used to build a model (Train) and a separate set used to honestly evaluate it (Test), so performance is not judged on the same data the model was fitted to.
Cross-validation (CV)	Repeatedly holding out part of the Training data, fitting on the rest, and predicting the held-out part, to get a more realistic performance estimate than fitting and testing on the exact same data.
Leave-one-out (LOO)	A cross-validation strategy where each single sample is held out and predicted once.
Stratified k-fold	Cross-validation folds built so each fold preserves the overall class/response proportions.
PCA (Principal Component Analysis)	An unsupervised technique that summarises many correlated variables into a smaller number of uncorrelated "components" capturing most of the variation in the data.
Scores	The coordinates of each sample in a reduced-dimension space (e.g. PCA/PLS components).
Loadings	How strongly each original variable contributes to a given component/latent variable.
Explained variance	The percentage of the total information (variance) in the data captured by a given component.
Hotelling's T^2	A statistic measuring how extreme a sample is <i>within</i> a model (large values = unusual combination of scores).
Q residual (SPE)	A statistic measuring how poorly a sample is explained/reconstructed <i>by</i> a model (large values = the model does not fit this sample well).

Term	Practical meaning
HCA (Hierarchical Cluster Analysis)	An unsupervised technique that groups samples into a tree (dendrogram) based on similarity.
Dendrogram	A tree diagram showing the order and distance at which samples/clusters merge in HCA.
Silhouette score	A per-sample measure of how well it fits its assigned cluster compared to the next-closest cluster; higher/positive is better.
K-means	An unsupervised technique that partitions samples into a fixed number (k) of clusters around estimated centroids.
t-SNE	A non-linear technique for visualising high-dimensional data in 2D/3D, preserving local neighbourhood structure (not global distances).
SIMCA	Soft Independent Modelling of Class Analogies: a one-class modelling technique that accepts or rejects new samples based on whether they fit a single target class's own PCA-based envelope.
LDA / QDA (Linear / Quadratic Discriminant Analysis)	Classification methods that separate classes using straight (LDA) or curved (QDA) boundaries based on each class's estimated distribution.
PLS-DA	Partial Least Squares Discriminant Analysis: a classification method built on PLS latent variables chosen to separate classes.
KNN (K-Nearest Neighbours)	A classification method that assigns a sample to the majority class among its k closest training samples.
VIP (Variable Importance in Projection)	A score (PLS/PLS-DA) summarising how much a variable contributes across all retained latent variables; larger VIP = more influential variable.
MLR (Multiple Linear Regression / OLS)	A regression method fitting a straight-line relationship between all predictors and the response simultaneously.
PCR (Principal Component Regression)	Regression performed on PCA scores instead of the raw predictors, to handle many correlated variables.
PLS (Partial Least Squares)	Regression (or, as PLS-DA, classification) performed on latent variables chosen to correlate with the response, handling many correlated predictors.
Latent variable (LV) / Component	A constructed variable (a combination of the original predictors) used by PCA/PCR/PLS/PLS-DA instead of the raw variables.
RMSE (Root Mean Squared Error)	Average prediction error of a regression model, in the same units as the response; lower is better.
R^2 (Coefficient of determination)	The proportion of variation in the response explained by the model (0–1, or as a %); higher is better.
Accuracy / Sensitivity / Specificity	Classification performance measures: overall correct-rate, correct-rate within the positive class, and correct-rate within the negative class, respectively.
Confusion matrix	A table of actual vs. predicted classes, showing exactly which classes are being confused with which.
FAR / FRR	False Acceptance Rate / False Rejection Rate: in one-class modelling (SIMCA), the rate of wrongly accepting non-target samples / wrongly rejecting target samples.

Term	Practical meaning
p-value	The probability of seeing a result at least as extreme as observed if there were truly no effect; conventionally, $p < 0.05$ is treated as "statistically significant".
Effect size	A measure of <i>how large</i> a difference or association is, independent of sample size – always worth checking alongside a p-value.
Savitzky-Golay smoothing/derivative	A spectral preprocessing technique that locally fits a polynomial to smooth a spectrum or estimate its derivative.
SNV (Standard Normal Variate)	A row-wise spectral preprocessing step correcting scatter effects by centring/scaling each individual spectrum.
MSC (Multiplicative Scatter Correction)	A row-wise spectral preprocessing step correcting scatter effects relative to a reference spectrum.
Baseline correction (ALS)	A preprocessing step that estimates and removes a slowly varying background/drift from each spectrum.
Autoscaling	Centring and scaling a variable to zero mean and unit variance (a "z-score").
Pareto scaling	Centring a variable and dividing by the square root of its standard deviation – a compromise between no scaling and full autoscaling.

19.2 Where to get help

- Application-specific problems: Chapter 18.
- Feedback / bug reports: the feedback link on the Credits tab (Chapter 15).

⚠️ TODO

This appendix reflects the modules and options present in the application at the time this guide was written. If a module is added, removed, or renamed in a future version, update the corresponding chapter under `docs/modules/` and this glossary together.

Chapter 20

Third-Party Libraries

Cactus is built in R and Shiny, and relies on a number of open-source packages to provide its interface, charts, and statistical/chemometric methods. This appendix lists every package loaded by the application (as declared in `global.R`), grouped by the part of the workflow it supports, with a link to its official documentation.

Note

These are development dependencies, not something you need to install or configure yourself – they are bundled with the application (Chapter 2). This page is provided for transparency and for anyone who wants to read more about the methods behind a given feature.

20.1 Interface (core UI)

Package	Used for
shiny	The web application framework Cactus is built on.
shinydashboard	The dashboard layout (sidebar, header, boxes).
shinydashboardPlus	Extra dashboard components (the right-hand settings panel).
shinyWidgets	Extra input controls (pickers, switches, action buttons) used throughout the app.
shinyjs	Show/hide interface elements and run small JavaScript actions from R.
DT	Interactive, searchable data tables.
plotly	All interactive charts (zoom, pan, hover, export) – see Chapter 16.
htmlwidgets	The underlying framework that lets R render interactive widgets like plotly and DT in the browser.
jsonlite	Reading/writing JSON data (e.g. saved user preferences, login status).
RColorBrewer	The colour palettes offered in the Qualitative Palette picker.
colourpicker	The colour-picker input used for categorical colours.
curl	Network requests (used for the optional external login service).
shinydisconnect	The "connection lost" screen shown if the app disconnects.

20.2 Data import

Package	Used for
readxl	Reading Excel files (<code>.xls/.xlsx</code>).
data.table	Fast reading of CSV/TXT files.

20.3 Edit module

Package	Used for
lubridate	Detecting and parsing date/time columns.
dplyr	General data manipulation (Edit pipeline, Missing Values).
tibble	Modern data-table utilities used internally.
readr	Writing CSV files (e.g. the Edited Data download).
rlang	Internal programming utilities.

20.4 Split module

Package	Used for
withr	Makes random splits reproducible (the Seed setting).
prospectr	The Kennard-Stone and DUPLEX split methods.

20.5 Missing values

Package	Used for
recipes	The preprocessing/imputation pipeline framework (also used by Preprocessing and every modelling module).

20.6 Visualization

Package	Used for
scales	Mapping values to visual scales (e.g. Bubble Plot size).
MASS	The 2D Density Chart calculation.

Package	Used for
ggplot2	Building certain charts before they are made interactive.

20.7 Unsupervised modelling

Package	Used for
mdatools	PCA, and the underlying engine behind PLS and SIMCA.
reshape2	Reshaping data for the Loadings bar chart.
writexl	Excel export (e.g. "Save PCA data").
dendextend	Colouring the HCA dendrogram.
cluster	The Silhouette Plot calculation (K-means, HCA).
Rtsne	The t-SNE technique.

20.8 Classification

Package	Used for
discrim	The LDA and QDA engines.
parsnip	A common interface for defining models (shared by Classification and Regression).
workflows	Bundles a preprocessing recipe together with a model.
tune	Hyperparameter tuning (e.g. choosing the best k for KNN).
rsample	Builds the cross-validation folds (Section 11.2).
yardstick	Calculates model metrics (Accuracy, Sensitivity, RMSE, R^2 , etc.).
dials	Generates the tuning grids used when searching for the best model settings.
plsmod	The PLS-DA engine.
butcher	Shrinks a trained model before saving it, for smaller downloaded files.
kknn	The KNN engine.

20.9 Preprocessing

Package	Used for
shinyFeedback	Inline validation warnings (e.g. invalid Savitzky-Golay settings).
shinycustomloader	Loading spinners while a preprocessing preview is calculated.

Package	Used for
sortable	The drag-and-drop pipeline builder.

20.10 Regression

Package	Used for
mixOmics	The PLS and PLS-DA engine (Bioconductor package).
broom	Tidies model output for the MLR Coefficients/Model Statistics tables.
gvlma	The MLR Assumptions diagnostic test.
tidyr	Reshaping data for scores/loadings charts.
colorspace	Generates the colour palette used for loadings colour pickers.